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INTRODUCTION

Techniques for probing the immunophenotype and function of cells of the immune system are becoming ever more sophisticated. In addition, it is possible to identify the function of individual genes by mutation and the creation of transgenic and 'knockout' animals.

ISOLATION OF LEUKOCYTE SUBPOPULATIONS

Bulk techniques

Separation based on physical parameters

Separation of cells on the basis of their differential **sedimentation rate**, which roughly correlates with **size**, can be carried out by centrifugation through a density gradient. Cells can be increased in mass by selectively binding particles such as red cells to their surface, the most notable example being the rosettes formed when sheep erythrocytes bind to the CD2 of human T-cells.

Buoyant density is another useful parameter. Centrifugation of whole blood over isotonic Ficoll-Hypaque (sodium metrizoate) of density 1.077 g/ml leaves the mononuclear cells (lymphocytes, monocytes and natural killer (NK) cells) floating in a band at

the interface, while the erythrocytes and polymorphonuclear leukocytes, being denser, travel right down to the base of the tube. **Adherence** to plastic surfaces largely removes phagocytic cells, while passage down nylon-wool columns greatly enriches lymphocyte populations for T-cells at the expense of B-cells.

Separation exploiting biological parameters

Actively phagocytic cells which take up small iron particles can be manipulated by a magnet deployed externally. Lymphocytes which divide in response to a polyclonal activator (see p. 164), or specific antigen, can be eliminated by allowing them to incorporate 5-bromodeoxyuridine (BrdU); this renders them susceptible to the lethal effect of UV irradiation.

Selection by antibody

Several methods are available for the selection of cells specifically coated with antibody, some of which are illustrated in figure 7.1. Addition of complement or anti-Ig toxin conjugates will eliminate such populations. Magnetic beads coated with anti-Ig form clusters with antibody-coated cells which can be readily separated from uncoated cells. Another useful bulk selection technique is to pan antibody-coated cells on anti-Ig adsorbed to a surface. One variation on this theme used to isolate bone marrow stem cells with anti-CD34 is to coat the cells with biotinylated antibody and select with an avidin column or avidin magnetic beads.

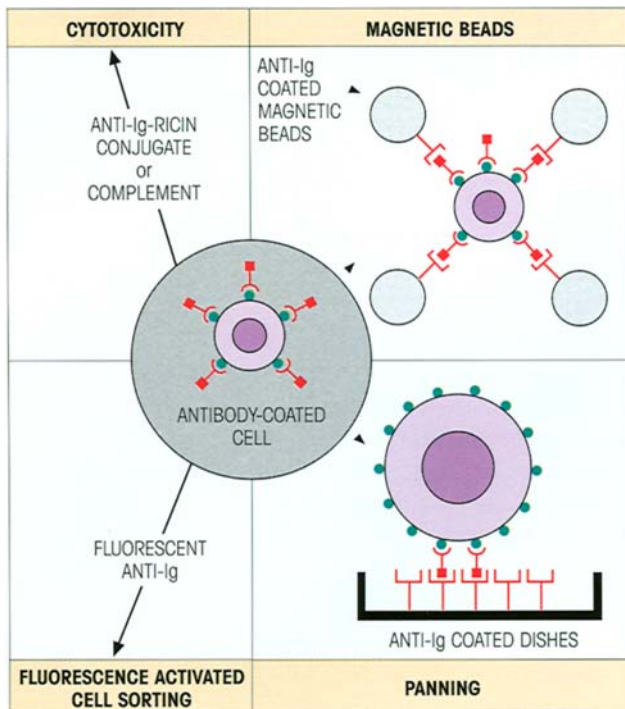


Figure 7.1. Major methods for separating cells coated with a specific antibody.

Cocktails of antibodies coated onto beads are used in cell separation columns for the depletion of specific populations leading to, for example, enriched CD4⁺ CD45RA⁻ or CD4⁺ CD45RO⁻ lymphocytes.

Cell selection by the FACS

Cells coated with fluorescent antibody can be separated in the fluorescence-activated cell sorter (FACS) as described in Milestone 7.1 and figure 7.2 (see more in-depth discussion under 'Flow cytofluorimetry', p. 133).

Enrichment of antigen-specific populations

Selective expansion of antigen-specific T-cells by repeated stimulation with antigen and presenting cells in culture, usually alternated with interleukin-2 (IL-2) treatment, leads to an enrichment of heterogeneous T-cells specific for different epitopes on the antigen. Such **T-cell lines** can be distributed in microtiter wells at a high enough dilution such that **on average** there is less than one cell per well; pushing the cells to proliferate with antigen or anti-CD3 produces single T-cell clones which can be maintained with much obsessional care and attention, but my goodness they can be a pain! Potentially immortal **T-cell hybridomas**, similar in

principle to B-cell hybridomas, can be established by fusing cell lines with a T-tumor line and cloning.

Animals populated essentially by a single T-cell specificity can be produced by introducing the T-cell receptor α and β genes from a T-cell clone, as a transgene (see below); since the genes are already rearranged, their presence in every developing T-cell will switch off any other V β gene recombinations.

No one has succeeded in cloning B-cells, except as hybridomas or Epstein-Barr virus-transformed cell lines, although, as with T-cells, transgenic animals expressing the same antibody in all their B-cells have been generated.

IMMUNOHISTOCHEMISTRY—LOCALIZATION OF ANTIGENS IN CELLS AND TISSUES

Immunofluorescence techniques

Because fluorescent dyes such as fluorescein and rhodamine can be coupled to antibodies without destroying their specificity, the conjugates can combine with antigen present in a tissue section and be visualized in the fluorescence microscope. In this way, the distribution of antigen throughout a tissue and within cells can be demonstrated. Looked at another way, the method can also be used for the detection of antibodies directed against antigens already known to be present in a given tissue section or cell preparation. There are two general ways in which the test is carried out.

Direct test with labeled antibody

The antibody to the tissue antigen is conjugated with the fluorochrome and applied directly (figure 7.3a). For example, suppose we wished to show the distribution of a thyroid autoantigen reacting with the autoantibodies present in the serum of a patient with Hashimoto's disease, a type of thyroid autoimmunity. We would isolate IgG from the patient's serum, conjugate it with fluorescein, and apply it to a section of human thyroid on a slide. When viewed in the fluorescence microscope we would see that the cytoplasm of the follicular epithelial cells was brightly stained (cf. figure 19.1a).

By using two antisera conjugated to dyes which emit fluorescence at different wavelengths, two different antigens can be identified simultaneously in the same preparation. In figure 2.6e, direct staining of fixed plasma cells with a mixture of rhodamine-labeled anti-IgG and fluorescein-conjugated anti-IgM craftily demonstrates that these two classes of antibody are produced by different cells. The technique of coupling

Milestone 7.1 — The Fluorescence-activated Cell Sorter (FACS)

The FACS was developed by the Herzenbergs and their colleagues to quantify the surface molecules on individual white cells by their reaction with fluorochrome-labeled monoclonal antibodies and to use the signals so generated to separate cells of defined phenotype from a heterogeneous mixture.

In this elegant but complex machine, the fluorescent cells are made to flow obediently in a single stream past a laser beam. Quantitative measurement of the fluorescent signal in a suitably placed photomultiplier tube relays a signal to the cell as it emerges in a single droplet; the cell

becomes charged and can be separated in an electric field (figure M7.1.1). Extra sophistication can be introduced by using additional lasers and fluorochromes, and both 90° and forward light scatter. This will be elaborated upon in the section on flow cytofluorimetry describing how this technique can be used for quantitative multi-parameter analysis of single-cell populations (cf. figure 7.6). Suffice to state that these latest FACS machines permit the isolation of cells with a complex phenotype from a heterogeneous population with a high degree of discrimination.

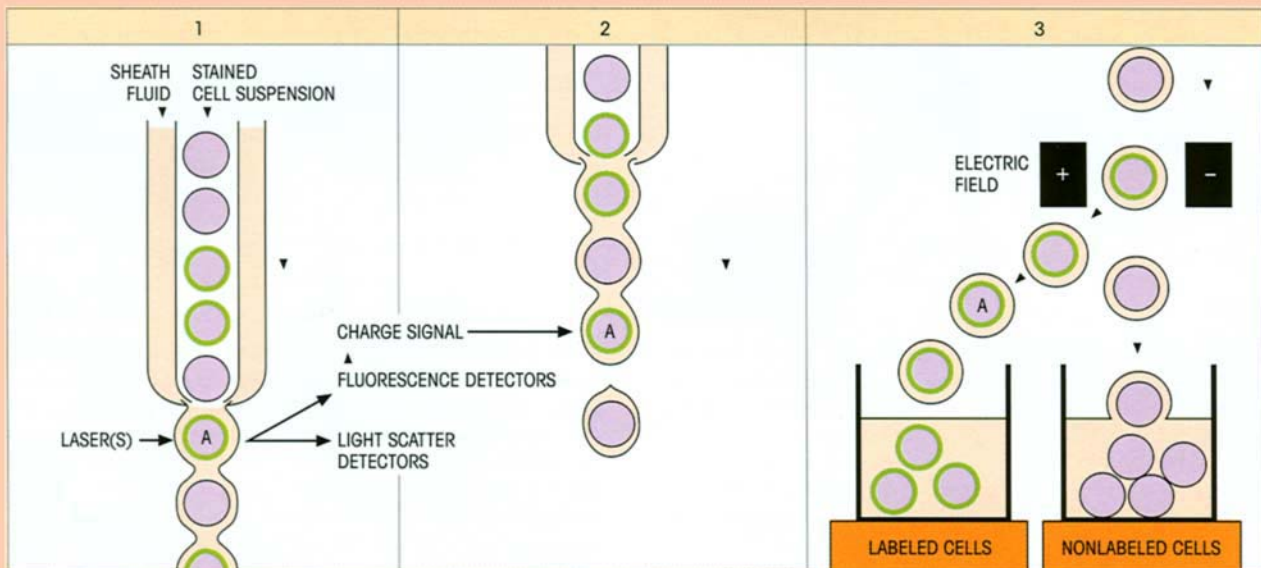


Figure M7.1.1. The principle of the FACS for flow cytofluorimetry of the fluorescence on stained cells (green rimmed circles) and physical separation from unstained cells. The charge signal

can be activated to separate cells of high from low fluorescence and, using light scatter, of large from small size and dead from living.

biotin to the antiserum and then finally staining with fluorescent avidin is often employed.

Indirect test for antibody

In this double-layer technique, the unlabeled antibody is applied directly to the tissue substrate and visualized by treatment with a fluorochrome-conjugated anti-immunoglobulin serum (figure 7.3b). In this case, in order to find out whether or not the serum of a patient has antibodies to thyroid epithelial cells, we would first treat a thyroid section with the serum, wash well and then apply a fluorescein-labeled rabbit

anti-human immunoglobulin; if antibodies were present, there would be staining of the thyroid epithelial cells.

This technique has several advantages. In the first place the fluorescence is brighter than with the direct test since several fluorescent anti-immunoglobulins bind on to each of the antibody molecules present in the first layer (figure 7.3b). Second, even when many sera have to be screened for specific antibodies it is only necessary to prepare (or, more usually the case, purchase) a single labeled reagent, viz. the anti-immunoglobulin. Furthermore, the method has great flexibility. For example, by using conjugates of antisera

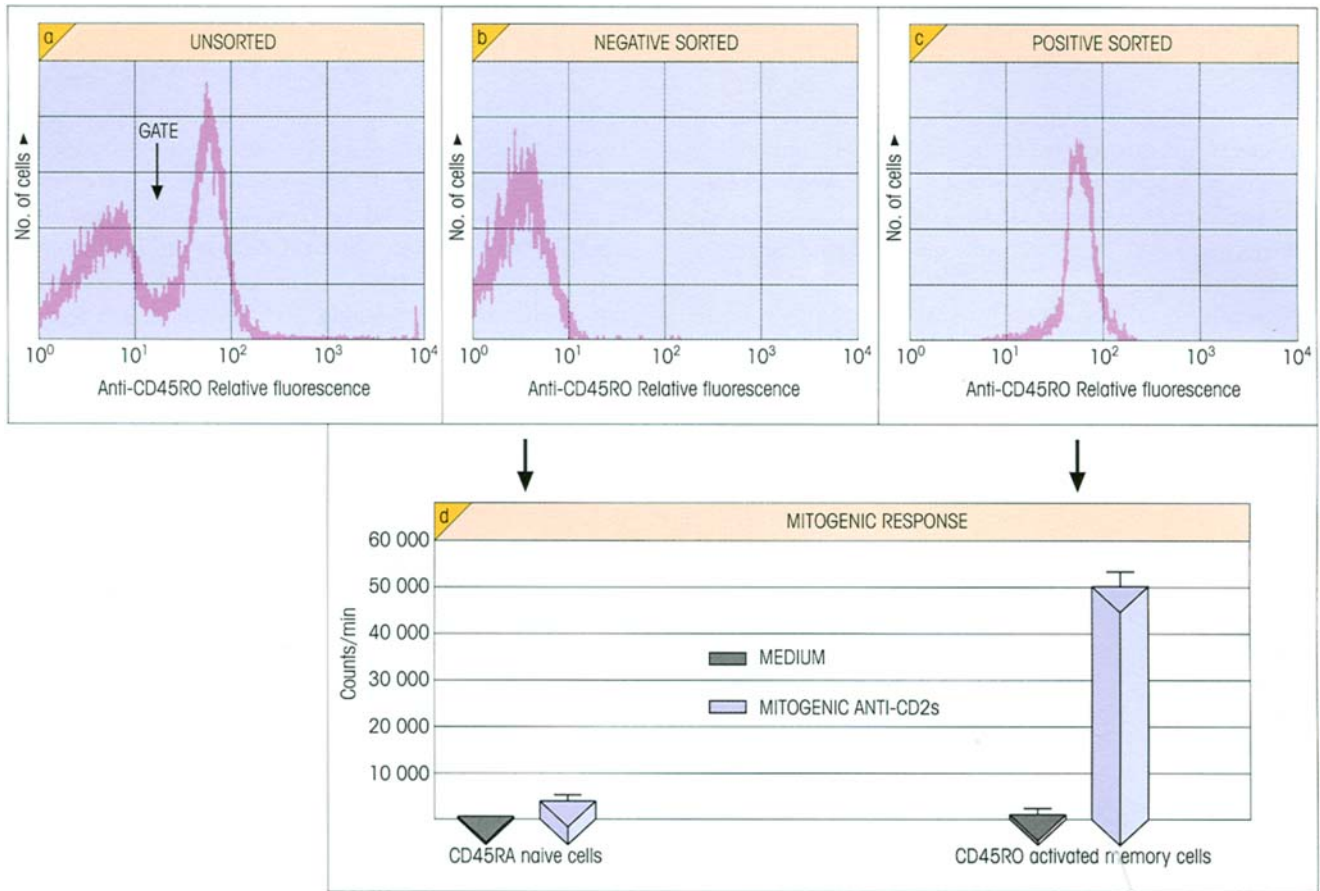


Figure 7.2. Separation of activated peripheral blood memory T-cells (CD45RO positive) from naive T-cells (CD45RO negative; but positive for the CD45RA isoform) in the FACS after staining the surface of the living cells in the cold with a fluorescent monoclonal antibody to the CD45RO (see p. 196). The unsorted cells showed two peaks (a); cells with fluorescence intensity lower than the arbitrary gate were separated from those with higher intensity giving

(b) negative (CD45RA) and (c) positive (CD45RO) populations, which were each tested for their proliferative response to a mixture of two anti-CD2 monoclonals (OKT11 and GT2) in the presence of 10% antigen-presenting cells (d). ^3H -Thymidine was added after 3 days and the cells counted after 15h. Clearly the memory cell population proliferated, whereas the naive population did not. (Data kindly provided by D. Wallace and R. Hicks.)

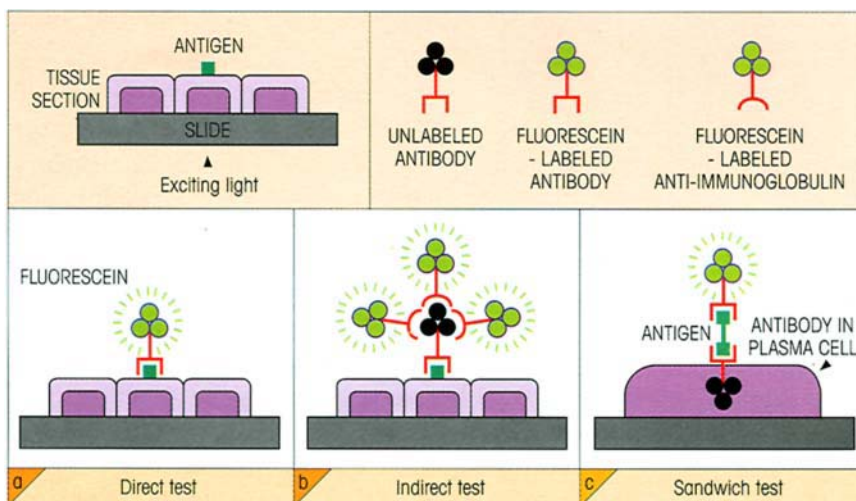


Figure 7.3. The basis of fluorescence antibody tests for identification of tissue antigens or their antibodies. ● = fluorescein labeled.

to individual immunoglobulin heavy chains, the distribution of antibodies among the various classes and subclasses can be assessed at least semiquantitatively. One can also test for complement fixation on the tissue section by adding a mixture of the first antibody plus a source of complement, followed by a fluorescent anticomplement reagent as the second layer.

Further applications of the indirect test may be seen in Chapter 19.

High resolution with the confocal microscope

Fluorescent images at high magnification are usually difficult to resolve because of the flare from slightly out of focus planes above and below that of the object. All that is now past, with the advent of commercially available **scanning confocal microscopes** which focus the image of the laser source in a fine plane within the cell and collect the fluorescence emission in a photomultiplier tube (PMT) with a confocal aperture. Fluorescence from planes above and below the object plane fails to reach the PMT and so the sharpness of the image is unaffected. An X–Y scanning unit enables the whole of the specimen plane to be interrogated *quantitatively* and, with suitable optics, two different fluorochromes can be used simultaneously. The instrument software can compute three-dimensional fluorescent images from an automatic series of such X–Y scans accumulated in the Z axis (figure 7.4) and rotate them at the whim of the operator.

Flow cytofluorimetry

Cell surface antigens can be detected and localized by the use of labeled antibodies. Because antibodies cannot readily penetrate living cells except by endocytosis, treatment of cells with labeled antibody in the cold (to minimize endocytosis) leads to staining only of antigens on the surface (figure 7.2). Remember also a previous example of fluorescent staining of the surface of B-lymphocytes with anti-immunoglobulin (figure 2.6c).

It is possible to stain single cells with several different fluorochromes (figure 7.5) and analyse the cells in individual droplets as they flow past the monitoring section of a **flow cytometer**. This piece of equipment is essentially a FACS machine without the cell-sorting facility (figure 7.6). These instruments record quantitative data relating to the surface antigen content and physical nature of each individual cell, with multiple parameters being assessed per cell to give a phenotypic analysis on a single cell rather than a population average (figure 7.7). With the impressive number of

monoclonal antibodies and of fluorochromes to hand, highly detailed analyses are now feasible, with a notable contribution to the diagnosis of leukemia (cf. p. 382).

We can also probe the cell *interior* in several ways. Permeabilization to allow penetration by fluorescent antibodies (preferably with small Fab or even single-chain Fv fragments) gives a readout of cytokines and other intracellular proteins. Cell cycle analysis can be achieved with DNA-binding dyes such as propidium iodide to measure DNA content and antibody detection of BrdU incorporation to visualize DNA synthesis. In addition, fluorescent probes for intracellular pH, thiol concentration, Ca^{2+} , Mg^{2+} and Na^{+} have been developed.

A powerful analytical tool is the measurement of the intracellular expression of artificially introduced genes which comprise a coding sequence or regulatory region of interest fused to another gene for a molecule, such as green fluorescent protein (GFP), luciferase or LacZ (*Escherichia coli* β -galactosidase, detected via hydrolysis of the fluorogenic substrate fluorescein digalactoside). Depending on the construct used, the gene can be utilized as a marker for developmental and migration studies or as a reporter under the control of specified promoter or enhancer elements. As an example of the fancy flow cytofluorimetric analyses that can be carried out, transgenic replacement (cf. p. 143) of endogenous RAG-2 genes with a RAG-2–GFP fusion gene permitted the study of RAG-2 expression in individual lymphocytes whose phenotype was simultaneously defined by monoclonal antibody staining.

Other labeled antibody methods

In place of fluorescent markers, enzymes such as alkaline phosphatase (cf. figure 18.7) or peroxidase can be coupled to antibodies and then visualized by conventional histochemical methods at both light microscope (cf. figure 8.7f) and electron microscope level.

Colloidal gold bound to antibody is being widely used as an electron-dense immunolabel by electron microscopists. At least three different antibodies can be applied to the same section by labeling them with gold particles of different size (cf. figure 9.13). A new ultra-small probe consisting of Fab' fragments linked to undecagold clusters allows more accurate spatial localization of antigens and its small size enables it to mark sites which are inaccessible to the larger immunolabels. However, clear visualization requires a high-resolution scanning transmission electron microscope.

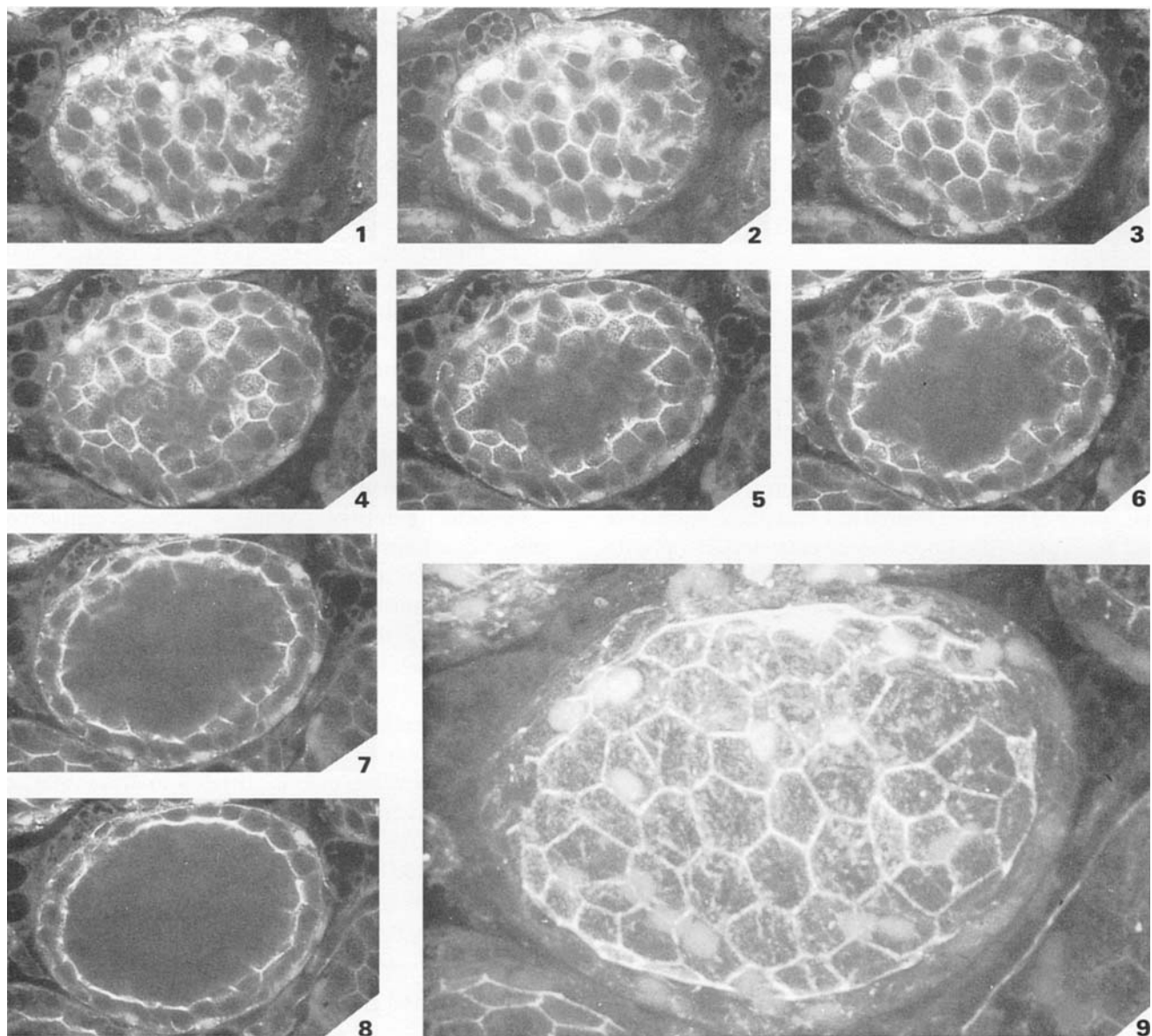


Figure 7.4. Construction of a three-dimensional fluorescent image with the confocal microscope. A spherical thyroid follicle in a thick razor-blade section of rat thyroid fixed in formalin was stained with a rhodamine–phalloidin conjugate which binds F-actin (similar results obtained with antibody conjugates). Although the sample was very thick, the microscope was focused on successive planes at 1 μm intervals from the top of the follicle (image no. 1) to halfway through (image no. 8), the total of the images representing a hemisphere. Note how the fluorescence in one plane does not interfere with that

in another and that the composite photograph (image no. 9) of images 1–8 shows all the fluorescent staining in focus throughout the depth of the hemisphere. Clearly the antibody is staining hexagonal structures close to the apical (inner) surface of the follicular epithelial cells. Erythrocytes are visible near the top of the follicle. (Negatives kindly supplied by Dr Anna Smallcombe were taken by Bio-Rad staff on a Bio-Rad MRC-600 confocal imaging system using material provided by Professor V. Herzog and Fr. Brix of Bonn University.)

GENE EXPRESSION

Messenger RNA can be extracted from cells or tissues and then analysed in dot blots, by northern blotting or by PCR. The development of microarray technologies now permits the simultaneous measurement of expression of thousands of genes in a single experiment. Oligonucleotides or cDNA fragments are robotically

spotted onto a gene chip and cDNA generated from, for example, T-cell mRNA is labeled and hybridized to the genes on the microarray. This provides a quantitative comparison of expression for every gene present on the chip. By accumulating such data it is possible to build up a complete picture of which genes are expressed in which cells (figure 7.8). One area in which this technology is being rapidly deployed is in the

analysis of differences in gene expression between a tumor cell and its normal counterpart, thereby illuminating possible targets for therapeutic intervention.

Localization of mRNA to individual cells can be obtained by *in situ* hybridization of labeled oligonucleotide probes to either cell smears or to tissue sections.

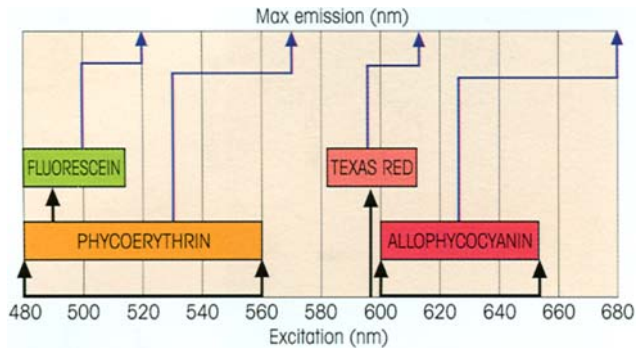


Figure 7.5. Fluorescent labels used in flow cytometry. The fluorescein longer wave emission overlaps with that of Texas Red and is corrected for in the software. The phycobiliproteins of red algae and cyanobacteria effect energy transfer of blue light to chlorophyll for photosynthesis; each molecule has many fluorescent groups giving a broad excitation range, but fluorescence is emitted within a narrow wavelength band with such high quantum efficiency as to obviate the need for a second amplifying antibody.

ASSESSMENT OF FUNCTIONAL ACTIVITY

The activity of phagocytic cells

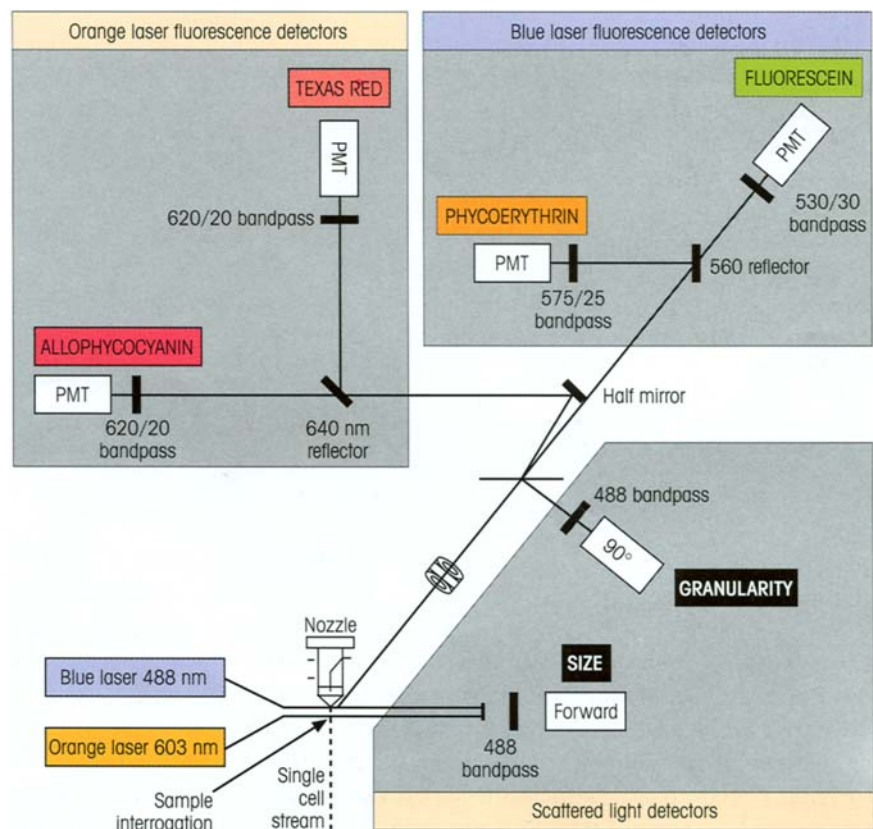
The major tests employed to assess neutrophil function are summarized in table 7.1.

Lymphocyte responsiveness

When lymphocytes are stimulated by antigen or polyclonal activators *in vitro* they usually undergo mitosis (cf. figure 2.6b) and release cytokines. Mitosis is normally assessed by the incorporation of radiolabeled ^3H -thymidine or ^{125}I -UdR (5-iododeoxyuridine) into the DNA of the dividing cells.

Cytokines released into the culture medium can be measured by immunoassay or by a bioassay using a cell line dependent on a particular cytokine for its growth and survival. Individual cells synthesizing cytokines can be enumerated in the flow cytometer by permeabilizing and staining intracellularly with labeled antibody; alternatively the ELISPOT technique (see below) can be applied. As usual, molecular biology has a valuable, if more sophisticated, input since T-cells transfected with an IL-2 enhancer-*lacZ* construct will switch on *lacZ* β -galactosidase expression on activation of the IL-2 cytokine response (cf.

Figure 7.6. Six-parameter flow cytometry optical system for multicolor immunofluorescence analysis. Cell fluorescence excited by the blue laser is divided into green (fluorescein) and orange (phycoerythrin) signals, while fluorescence excited by the orange laser is reflected by a mirror and divided into near red (Texas Red) and far red (allophycocyanin) signals. Blue light scattered at small forward angles and at 90° is also measured in this system, providing information on cell size and internal granularity respectively. PMT, photomultiplier tube. (Based closely on Hardy R.R. (1998) In Delves P.J. & Roitt I.M. (eds) *Encyclopedia of Immunology*, 2nd edn, p. 946. Academic Press, London.) The recent use of three lasers and nine different fluorochromes pushes the system even further, providing 11 parameters!



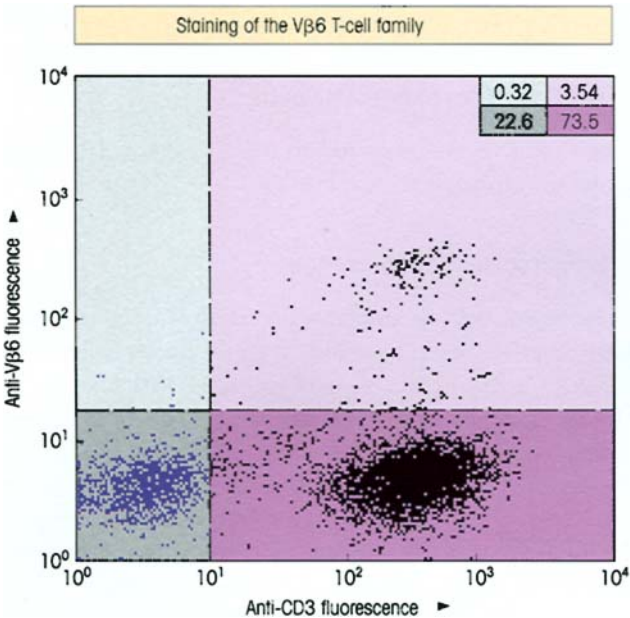


Figure 7.7. Cytofluorimetric analysis of TCR Vβ6 usage by human peripheral blood lymphocytes. Cells stained with fluoresceinated anti-TCR Vβ6 and phycoerythrin-conjugated anti-CD3. Each dot represents an individual lymphocyte and the numbers refer to the percentage of lymphocytes lying within the four quadrants formed by the two gating levels arbitrarily used to segregate positive from negative values. Virtually no lymphocytes bearing the T-cell receptors belonging to the Vβ6 family lack CD3, while 4.6% (3.5 out of 77.0) of the mature T-cells express Vβ6. (Data kindly provided by D. Morrison.)

Table 7.1. Evaluation of neutrophil function.

Function	Test
Phagocytosis	Measure the uptake of particles such as latex or bacteria by counting or by chemiluminescence
Respiratory burst	Measure reduction of nitroblue tetrazolium
Intracellular killing	Microbicidal test using viable <i>Staphylococcus aureus</i>
Directional migration	Movement through filters up concentration gradient of chemotactic agent such as formyl. Met. Leu. Phe
Surface LFA-1 and CR3 upregulation	Ascertained with monoclonal antibody staining

p. 167) and this can be readily revealed with a fluorescent or chromogenic enzyme substrate.

The ability of cytotoxic T-cells to kill their cell targets extracellularly is usually evaluated by a chromium release assay. Target cells are labeled with ⁵¹Cr and the release of radioactive protein into the medium over and above that seen in the controls is the index of cytotoxicity. The test is repeated at different ratios of effector to target cells. A similar technique is used to

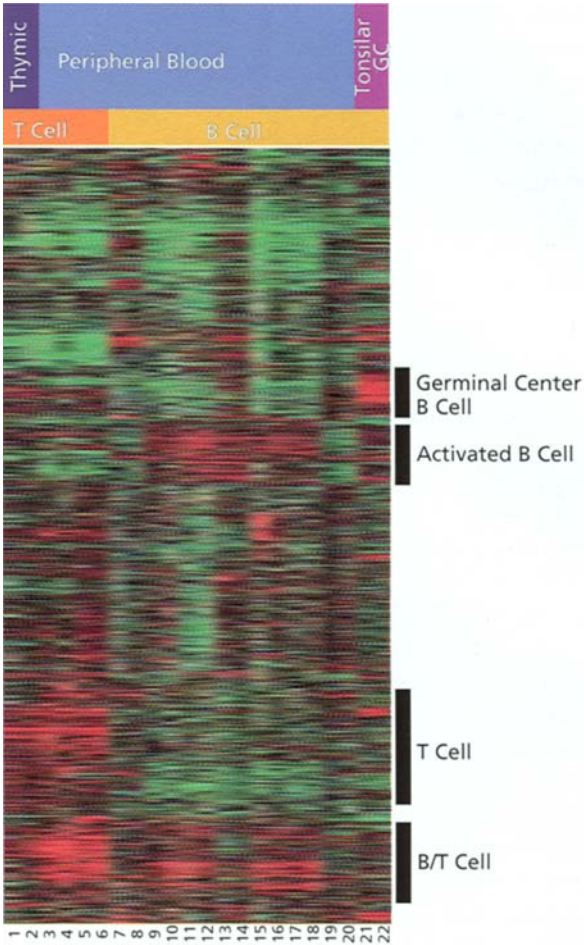


Figure 7.8. Gene expression during lymphocyte development and activation. The data were generated from over 3.8 million measurements of gene expression made on 13 637 genes using 243 microarrays. Each experiment represents a different cell population. For example, experiment 1 utilized polyclonally activated fetal CD4⁺ thymic cells, whereas experiment 2 shows the same population prior to stimulation. Overexpressed or induced genes are colored red, underexpressed or repressed genes green. Certain gene expression signatures become apparent in the different cell populations, indicated on the right. For example, the T-cell gene expression signature includes CD2, TCR, TCR signaling molecules and many cytokines. (Reproduced with permission of the authors and the publishers from Alizadeh A.A. & Staudt L.M. (2000) *Current Opinion in Immunology* 12, 219.)

measure extracellular killing of antibody-coated or uncoated targets by NK cells. Now a word of caution regarding the interpretation of *in vitro* assays. Since one can manipulate the culture conditions within wide limits, it is possible to achieve a result that might not be attainable *in vivo*. Let us illustrate this point by reference to cytotoxicity for murine cells infected with lymphocytic choriomeningitis virus (LCMV) or vesicular stomatitis virus (VSV). The most sensitive *in vitro* tech-

nique proved to be chromium release from target cells after secondary stimulation of the lymphocytes. However, this needs 5 days, during which time a relatively small number of memory CD8 cytotoxic T-cell precursors can replicate and surpass the threshold required to produce a measurable assay. Nonetheless, a weak cytotoxicity assay under these conditions was not reflected by any of the *in vivo* assessments of antiviral function implying that they had no biological relevance.

Apoptosis

Programed cell death occurs frequently in the immune system in a number of different situations and a variety of approaches can be used to measure the apoptotic cells. Gel electrophoresis will detect the characteristic laddering pattern of the DNA fragments generated due to internucleosomal cleavage (see figure 1.19). An alternative way to detect this DNA fragmentation, known as the TUNEL (*TdT-mediated dUTP (deoxyuridine triphosphate) nick end labeling*) technique, utilizes the enzyme terminal deoxynucleotidyl transferase (*TdT*) to add labeled nucleotides to the 3' ends of the fragments. An earlier event in apoptosis is the loss of membrane symmetry resulting in the expression of phosphatidylserine on the cell surface—readily detected by flow cytometry using a labeled version of its ligand Annexin V. For sophisticated, more detailed analyses can be undertaken in which individual components of the apoptotic pathways, such as Bcl-2, Bcl-x_L, Bax, PARP (poly(ADP-ribose) polymerase) or specific procaspases and caspases, are measured in fluorimetric or colorimetric assays.

Precursor frequency

The magnitude of lymphocyte responses in culture is closely related to the number of antigen-specific lymphocytes capable of responding. Because of the clonality of the responses, it is possible to estimate the frequency of these antigen-specific precursors by **limiting dilution analysis**. In essence, the method depends upon the fact that, if one takes several replicate aliquots of a given cell suspension which would be expected to contain *on average* one precursor per aliquot, then Poisson distribution analysis shows that 37% of the aliquots will contain *no* precursor cells (through the randomness of the sampling). Thus, if aliquots are made from a series of dilutions of a cell suspension and incubated under conditions which allow the precursors to mature and be recognized through some amplification scheme, the dilution at which 37% of the aliquots give negative responses will be known to con-

tain an average of one precursor cell per aliquot, and one can therefore calculate the precursor frequency in the original cell suspension. An example is shown in some detail in figure 7.9.

It has been argued that limiting dilution analysis often underestimates the true precursor frequency. An accurate measure of the percentage of lymphocytes bearing a specific antigen receptor can be obtained by flow cytometry of cells stained with labeled antigen. In the case of B-cells this is fairly straightforward given that their antigen receptors recognize native antigen. However, it is only recently that technical finesse, in the form of peptide–MHC tetramers, has brought this technique to T-cells (figure 7.10). This approach overcomes the problem of the relatively weak intrinsic affinity of TCR for peptide–MHC by presenting a tagged peptide–MHC as a multivalent tetramer, thereby exploiting the bonus effect of multivalency (cf. p. 90). Peptide–MHC complexes are produced by permitting recombinant MHC molecules to refold with the appropriate synthetic peptide. The recombinant MHC molecules are biotinylated on a special carboxy-terminal extension, which ensures that the biotin is incorporated at a distance from the site to which the TCR binds, and mixed with fluorescently labeled strept-avidin, which not only binds biotin with a very high affinity but also has a valency of four with respect to the biotin—hence the formation of tetramers.

Numerous adaptations of this technology are appearing. For example, incubation at 37°C of tetramers bound to their cognate TCR leads to internalization; by tagging them with a toxin individual T-lymphocytes of a single specificity can be eliminated. Another approach is to use the FACS to directly sort stained cells into an ELISPOT microtiter plate in which cytokine secretion is measured, providing a functional analysis of the cells.

Enumeration of antibody-forming cells

The immunofluorescence sandwich test

This is a double-layer procedure designed to visualize specific intracellular antibody. If, for example, we wished to see how many cells in a preparation of lymphoid tissue were synthesizing antibody to *Pneumococcus* polysaccharide, we would first fix the cells with ethanol to prevent the antibody being washed away during the test, and then treat with a solution of the polysaccharide antigen. After washing, a fluorescein-labeled antibody to the polysaccharide would then be added to locate those cells which had specifically bound the antigen.

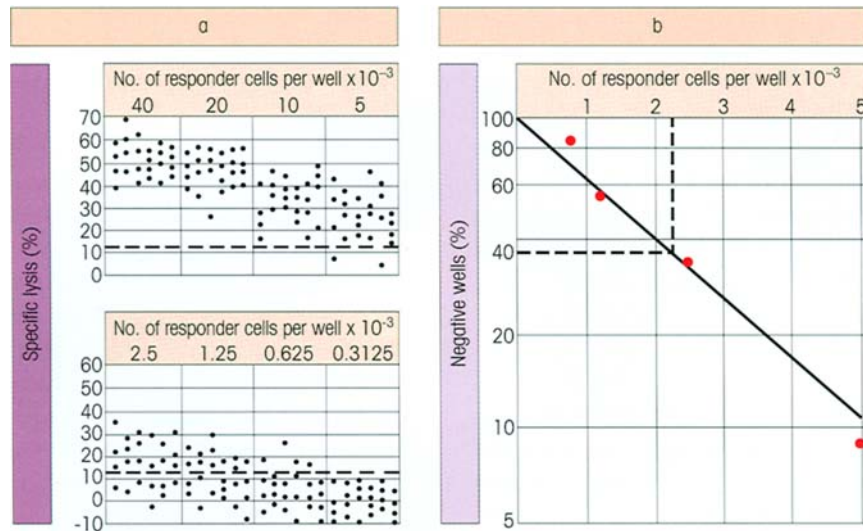


Figure 7.9. Limiting dilution analysis of cytotoxic T-cell precursor frequency in spleen cells from a BALB/c mouse stimulated with irradiated C57BL/6 spleen cells as antigen. BALB/c splenic responder cells were set up in 24 replicates at each concentration tested together with antigen and an excess of T-helper factors. The generation of cytotoxicity in each well was looked for by adding ^{51}Cr -labeled tumor cells (EL-4) of the C57BL/6 haplotype; cytotoxicity was then revealed by measuring the release of soluble ^{51}Cr -labeled intracellular material into the medium. (a) The points show the percentage of specific lysis of individual wells. The dashed line indicates three standard deviations above the medium release control,

and each point above that line is counted as positive for cytotoxicity. (b) The data replotted in terms of the percentage of negative wells at each concentration of responder cells over the range in which the data titrated (5×10^{-3} /well to 0.625×10^{-3} /well). The dashed line is drawn at 37% negative wells and this intersects the regression line to give a precursor (T_{cp}) frequency of 1 in 2327 responder cells. The regression line has an r^2 value of 1.00 in this experiment. (Reproduced with permission from Simpson E. & Chandler P. (1986) In Weir D.M. (ed.) *Handbook of Experimental Immunology*, figure 68.2. Blackwell Scientific Publications, Oxford.)

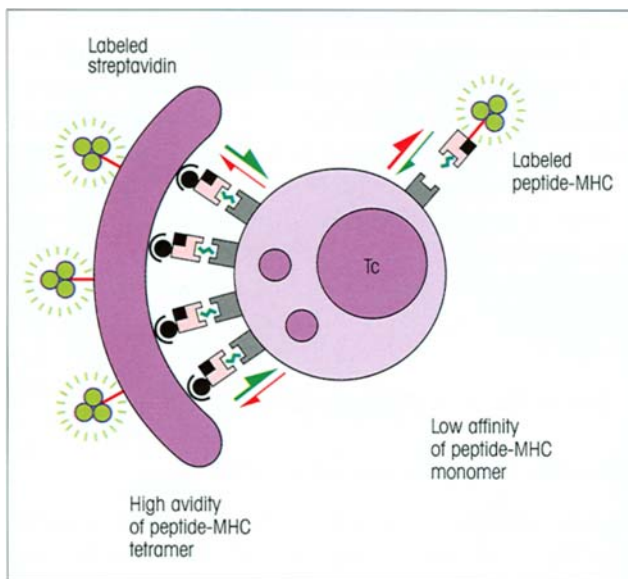


Figure 7.10. Peptide-MHC tetramer. A single fluorochrome-labeled peptide-MHC complex (top right) has only a low affinity for the TCR and therefore provides a very insensitive probe for its cognate receptor. However, by biotinylating (●) the MHC molecules and then mixing them with streptavidin, which has a valency of four with respect to biotin binding, a tetrameric complex is formed which has a much higher functional affinity (avidity) when used as a probe for the specific TCRs on the T-cell surface.

The name of the test derives from the fact that antigen is sandwiched between the antibody present in the cell substrate and that added as the second layer (figure 7.3c).

Plaque techniques

Antibody-secreting cells can be counted by diluting them in an environment in which the antibody formed by each individual cell produces a readily observable effect. In one technique, developed from the original method of Jerne and Nordin, the cells from an animal immunized with sheep erythrocytes are suspended together with an excess of sheep red cells and complement within a shallow chamber formed between two microscope slides. On incubation, the antibody-forming cells release their immunoglobulin which coats the surrounding erythrocytes. The complement will then cause lysis of the coated cells and a **plaque** clear of red cells will be seen around each antibody-forming cell (figure 7.11). Direct plaques obtained in this way largely reveal IgM producers since this antibody has a high hemolytic efficiency. To demonstrate IgG synthesizing cells it is necessary to increase the complement binding of the erythrocyte-IgG antibody complex by adding a rabbit anti-IgG serum; the 'indi-

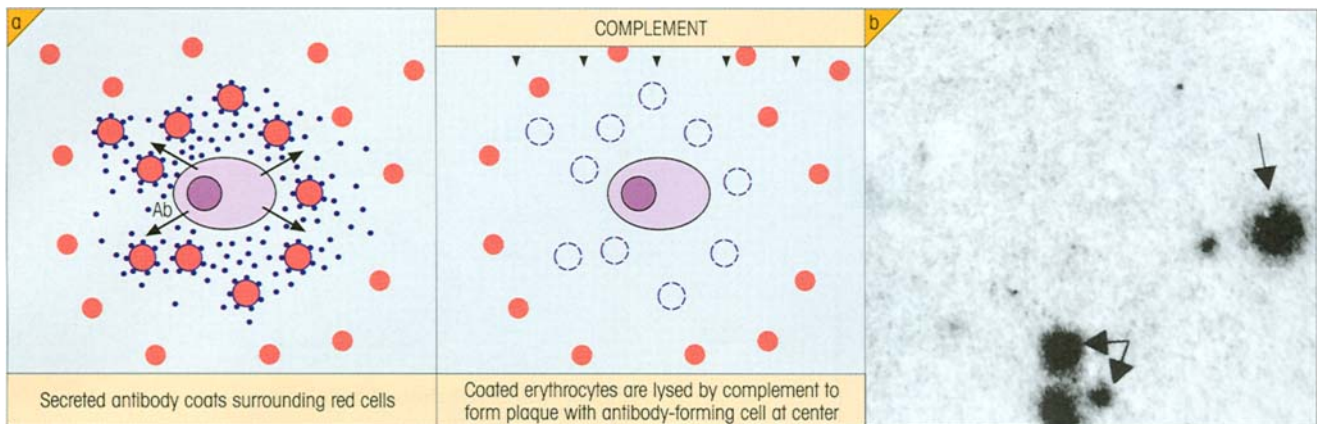


Figure 7.11. Jerne plaque technique for enumerating antibody-forming cells (Cunningham modification). (a) The *direct* technique for cells synthesizing IgM hemolysin is shown. The *indirect* technique for visualizing cells producing IgG hemolysins requires the addition of anti-IgG to the system. The difference between the plaques obtained by direct and indirect methods gives the number of 'IgG' plaques. The *reverse plaque* assay enumerates total Ig-producing

cells by capturing secreted Ig on red cells coated with anti-Ig. Multiple plaque assays can be carried out by a modification using microtiter plates. (b) Photograph of plaques which show as circular dark areas (some of which are arrowed) under dark-ground illumination. They vary in size depending upon the antibody affinity and the rate of secretion by the antibody-forming cell. (Courtesy of C. Shapland, P. Hutchings and Professor D. Male.)

rect plaques' thus developed can be used to enumerate cells making antibodies in different immunoglobulin subclasses, provided that the appropriate rabbit antisera are available. The method can be extended by coating an antigen such as *Pneumococcus* polysaccharide on to the red cell, or by coupling hapten groups to the erythrocyte surface.

In the **ELISPOT** modification, the antibody-forming cell suspension is incubated in microtiter wells containing filters coated with antigen. The secreted antibody is captured locally and is visualized, after removal of the cells, by treatment with enzyme-labeled anti-Ig and development of the color reaction with the substrate. The macroscopic spots can be readily enumerated (figure 7.12).

Analysis of functional activity by cellular reconstitution

Radiation chimeras

The entire populations of lymphocytes and polymorphs can be inactivated by appropriate doses of X-irradiation. Animals ablated in such a way may be reconstituted by injection of bone marrow hematopoietic stem cells which provide the precursors of all the formed elements of the blood (cf. figure 12.1). These chimeras of host plus hematopoietic grafted cells can be manipulated in many ways to analyse cellular function, such as the role of the thymus in the maturation of T-lymphocytes from bone marrow stem cells (figure 7.13).

Mice with severe combined immunodeficiency (SCID)

Mice with defects in the genes encoding the IL-2 receptor γ chain, the nucleotide salvage pathway enzymes adenosine deaminase or purine nucleoside phosphorylase, or the RAG enzymes, develop SCID due to a failure of B- and T-cells to differentiate. These special animals can be reconstituted with various human lymphoid tissues and their functions and responses analysed. Coimplantation of contiguous fragments of human fetal liver (hematopoietic stem cells) and thymus allows T-lymphopoiesis, production of B-cells and maintenance of colony-forming units of myeloid and erythroid lineages for 6–12 months. Adult peripheral blood cells injected into the peritoneal cavity of SCID mice treated with growth hormone can sustain the production of human B-cells and antibodies and can be used to generate human hybridomas making defined monoclonal antibodies. Immunotherapeutic antitumor responses can also be played with in these animals.

Cellular interactions in vitro

It is obvious that the methods outlined earlier for depletion, enrichment and isolation of individual cell populations enable the investigator to study cellular interactions through judicious recombinations. These interactions are usually more effective when the cells are operating within some sort of stromal network resembling the set-up of the tissues where their function is optimally expressed. For example, colonization of

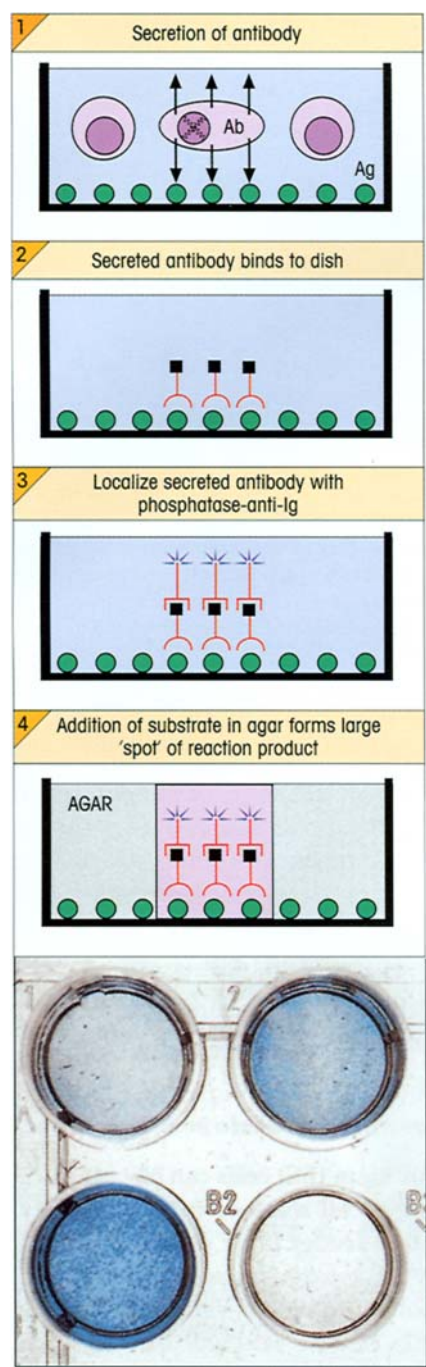


Figure 7.12. ELISPOT (from ELISA spot) system for enumerating antibody-forming cells. The picture shows spots formed by hybridoma cells making autoantibodies to thyroglobulin revealed by alkaline phosphatase-linked anti-Ig (courtesy of P. Hutchings). Increasing numbers of hybridoma cells were added to the top two and bottom left-hand wells which show corresponding increases in the number of 'ELISPOTS'. The bottom right-hand well is a control using a hybridoma of irrelevant specificity.

murine fetal thymus rudiments in culture with T-cell precursors enables one to follow the pattern of proliferation, maturation, TCR rearrangement and positive and negative selection normally seen *in vivo* (cf.

	Operation	Irradiation	Restitution	Induction of cell-mediated immunity
1	Sham thymectomy		Bone marrow	++
2	Thymectomy		Bone marrow	-
3	Thymectomy		Bone marrow + adult lymphocytes	++

Figure 7.13. Maturation of bone marrow stem cells under the influence of the thymus to become immunocompetent lymphocytes capable of cell-mediated immune reactions. X-irradiation (X) destroys the ability of host lymphocytes to mount a cellular immune response, but the stem cells in injected bone marrow can become immunocompetent and restore the response (1) unless the thymus is removed (2), in which case only already immunocompetent lymphocytes are effective (3). Incidentally, the bone marrow stem cells also restore the levels of other formed elements of the blood (red cells, platelets, neutrophils, monocytes) which otherwise fall dramatically after X-irradiation, and such therapy is crucial in cases where accidental or therapeutic exposure to X-rays or other anti-mitotic agents seriously damages the hematopoietic cells.

p. 225). An even more refined system involves the addition of selected lymphoid populations to disaggregated stromal cells derived from fetal thymic lobes depleted of endogenous lymphoid cells with deoxyguanosine. The cells can be spun into a pellet and cocultured in hanging drops; on transfer to normal organ culture conditions after a few hours, reaggregation to intact lobes takes place quite magically and the various differentiation and maturation processes then unfold.

Probing function with antibodies

Antibodies can be used to confirm the importance of cross-linking cell surface components for a number of functions. An excellent example is the induction of histamine release from mast cells by divalent $F(ab')_2$ anti-Fc ϵ RI but not by the univalent fragment. Similarly, divalent anti-Fc γ R triggers phagocytosis in macrophages, while a bispecific antibody to CD3 and CD4 brings the two molecules together on the T-cell surface and induces activation.

Spatially defined intracellular damage by chromophore-assisted laser inactivation can be carried out using antibodies. A neuronal growth cone in a living embryo, for example, can be permeabilized and loaded with anticalcineurin labeled with malachite green which absorbs light at 620 nm where most cellular constituents do not. A laser emitting light of this

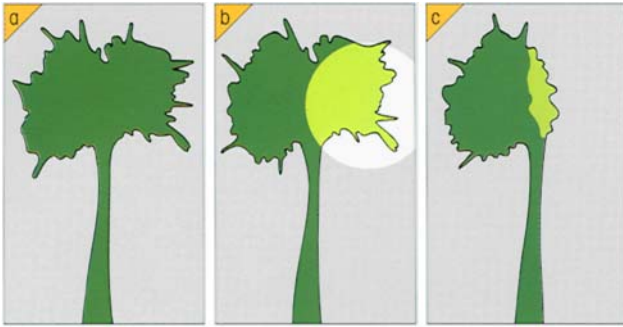


Figure 7.14. Microsurgery of a neuronal growth cone loaded with a malachite green-labeled antibody to calcineurin. The growth cone shows laser-induced retraction of membrane extensions: (a) before, (b) during and (c) after the laser flash. (Reproduced from a commentary by Muller B.K. & Bonhoeffer F. (1995) *Current Biology* 11, 1255 on experiments by Jay D.G. and colleagues (1995) *Nature* 376, 686, with permission.)

wavelength will irradiate the antibody–chromophore conjugate, so generating a pulse of highly reactive free radicals which cause inactivation in a small region of diameter around 15 Å. Figure 7.14 shows that the laser flash results in retraction of filopodial and lamellipodial membrane extensions. This in turn influences the direction of subsequent growth cone movement, so demonstrating the involvement of the calmodulin-dependent phosphatase, calcineurin, in growth cone steering. By operating at different embryonic stages the investigator can knock out functions in a temporally as well as spatially restricted fashion. This should prove to be a discriminating cellular microsurgical tool of considerable utility and applicability.

GENETIC ENGINEERING OF CELLS

Insertion and modification of genes in mammalian cells

Because gene transfer into mammalian cells is inefficient, it is customary to use immortal cell lines for such **transfections** and to include a selectable marker such as neomycin resistance. Genes can be introduced into cells as calcium phosphate precipitates or by electroporation, in which a brief electric pulse transiently creates holes in the cell membrane. Another approach is to incorporate the gene into liposomes which fuse with the cell membrane. Direct microinjection of DNA is also effective. Integration of the gene into the genome of a virus such as vaccinia provides an easy ride into the cell, although more stable long-term transfections are obtained with modified retroviral vectors. One of the latest fads is transfection by biolistics, the buzz word for biological ballistics. DNA coated on to gold

microparticles is literally fired from a high-pressure helium gun and penetrates the cells; even plant cells with their cellulose coats are easy meat for this technology. Skin and surgically exposed tissues can also be penetrated with ease.

Studying the effect of *adding* a gene, then, does not offer too many technological problems. How does one assess the impact of *removing* a gene? One versatile strategy to delete endogenous gene function is to target the gene's mRNA as distinct from the gene itself. Nucleotide sequences complementary to the mRNA of the target gene are introduced into the cell, usually in a form which allows them to replicate. The **antisense** molecules so produced base pair with the target mRNA and block translation into protein.

Introducing new genes into animals

Establishing 'designer mice' bearing new genes

Female mice are induced to superovulate and are then mated. The fertilized eggs are microinjected with the gene and surgically implanted in females. Between 5% and 40% of the implanted oocytes develop to term and, of these, 10–25% have copies of the injected gene, stably integrated into their chromosomes, detectable by PCR. These 'founder' transgenic animals are mated with nontransgenic mice and pure transgenic lines are eventually established (figure 7.15).

Expression of the transgene can be directed to particular tissues if the relevant promoter is included in the construct, for example the thyroglobulin promoter will confine expression to the thyroid. A different approach is to switch a gene on and off at will by incorporating an inducible promoter. Thus, the metallothionein promoter will enable expression of its linked gene only if zinc is added to the drinking water given to the mice. One needs to confirm that only the desired expression is obtained as, in some situations, promoters may misbehave leading to 'leaky' expression of the associated gene.

Transgenes introduced into embryonic stem cells

Embryonic stem (ES) cells can be obtained by culturing the inner cell mass of mouse blastocysts. After transfection with the appropriate gene, the transfected cells can be selected and reimplanted after injection into a new blastocyst. The resulting mice are chimeric, in that some cells carry the transgene and others do not. The same will be true of germ cells and, by breeding for germ-line transmission of the transgene, pure strains can be derived (figure 7.16).

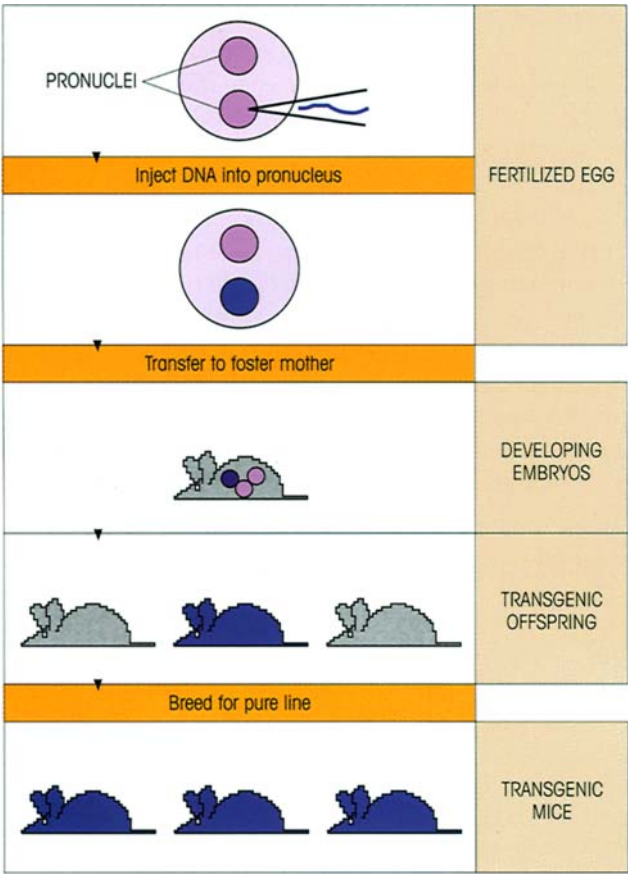


Figure 7.15. Production of pure strain transgenic mice by microinjection of fertilized egg, implantation into a foster mother and subsequent inbreeding.

The advantage over microinjection is that the cells can be selected after transfection, and this is especially important if **homologous recombination** is required in order to generate ‘**knockout mice**’ lacking the gene which has been targeted. In this case, a DNA sequence which will disrupt the reading frame of the endogenous gene is inserted into the ES cells. Because homologous recombination is a rare event compared to random integration, selectable markers are incorporated into the construct in order to transfer only those ES cells in which the endogenous gene has been deleted (figure 7.17). This is a truly powerful technology and the whole biological community has been suffused with boxing fever, knocking out genes right, left and center. Just a few examples of knockout mice of interest to immunologists are listed in table 7.2.

It is not a particularly rare finding to observe that knocking out a gene unexpectedly leads to developmental defects. Whilst this in itself can provide important information implicating the gene in the

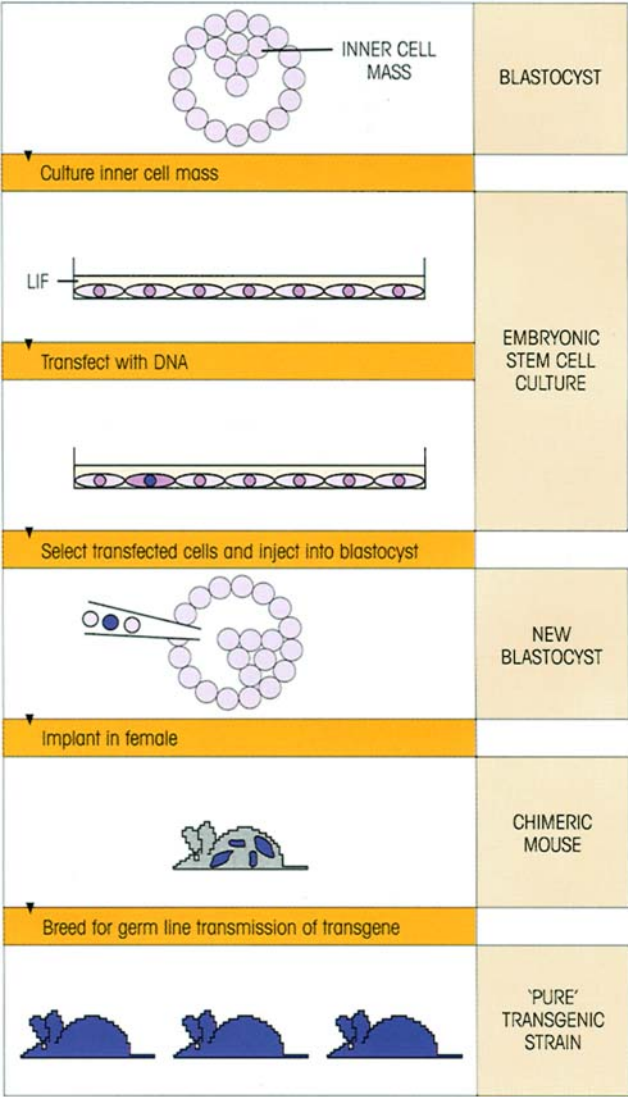


Figure 7.16. Introduction of a transgene through transfection of embryonic stem cells. The transfected cells can be selected, e.g. for homologous recombinant ‘knockouts’, before reimplantation. LIF, leukemia inhibitory factor.

developmental biology of the animal, it can frustrate the original aim of the experiment. Indeed, a number of knockouts are nonviable due to embryonic lethality. Never fear, ingenuity once again triumphs, in this case by the harnessing of viral or yeast recombinase systems. Instead of using a nonfunctional gene to create the knockout mouse, the targeting construct contains the normal form of the gene but flanked with recognition sequences (*loxP* sites) for a recombinase enzyme called Cre. These mice are mated with transgenic mice containing the bacteriophage P1-derived *Cre* transgene linked to an inducible or tissue-specific promoter. The endogenous gene of interest will be deleted only

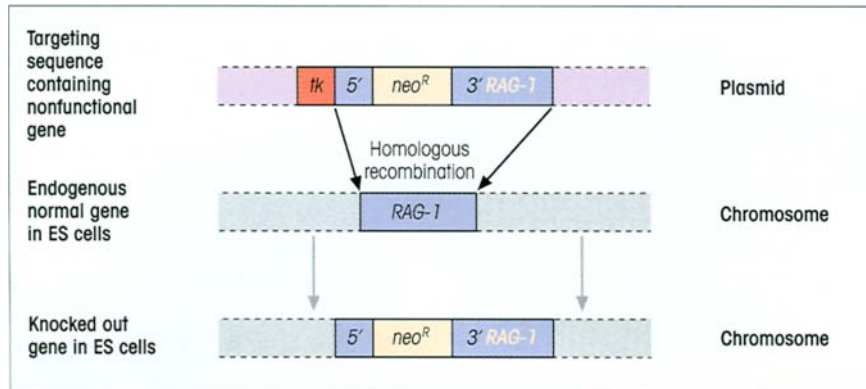


Figure 7.17. Gene disruption by homologous recombination with plasmid DNA containing a copy of the gene of interest (in this example *RAG-1*) into which a sequence specifying neomycin resistance (*neo^R*) has been inserted in such a way as to destroy the *RAG-1* reading frame between the 5' and 3' ends of the gene. Embryonic stem (ES) cells in which the targeting sequence has been incorporated into the chromosomal DNA by homologous recombination will be

resistant to the neomycin analog G418. Stem cells in which nonhomologous recombination into chromosomal DNA has occurred would additionally incorporate the *thymidine kinase* (*tk*) gene which can be used to destroy such cells by culturing them in the presence of ganciclovir, leaving only ES cells in which homologous recombination has been achieved. These are then used to create a knockout mouse.

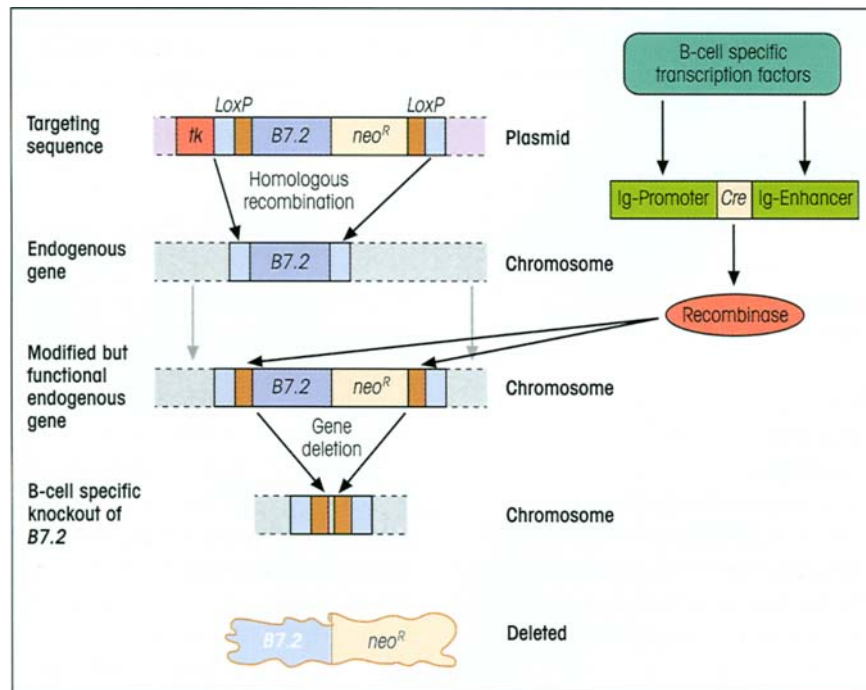


Figure 7.18. Conditional knockout. The endogenous gene that is under study (here *B7.2*) is homologously replaced in ES cells with an identical gene, as in figure 7.17, but here flanked by *loxP* sequences (brown boxes) and with the *neo^R* gene incorporated in a nondisruptive manner purely for selection purposes. Nonhomologous recombinants will contain the *tk* gene and are eliminated using ganciclovir. Transgenic animals are then generated from ES cells which are resistant to G418. If homozygous *B7.2-loxP* transgenics are mated with

mice which contain a transgene for the Cre recombinase under the control of specific regulatory elements, only those cells in which the promoter is active will produce the Cre enzyme necessary to delete the sequence flanked by *loxP*. The example given would represent an experiment aimed at investigating the effect of specifically knocking out *B7.2* in B-cells whilst maintaining its expression in, for example, dendritic cells.

when and where Cre is expressed (figure 7.18). The Cre/*loxP* system can also be organized in such a way as to turn on expression of a gene by incorporating a stop sequence flanked by *loxP* sites.

Mice in which an endogenous gene is purposefully

replaced by a functional gene, be it a modified version of the original gene or an entirely different gene, are referred to as '**knocked in mice**'. Hence, in the example above, knocking in a *loxP* flanked gene leads eventually to a knocked out gene in a selected cell type. Another

Table 7.2. Some gene 'knockouts' and their effects.

Knockout target	Phenotype of knockout mice
CD8 α -chain	Absence of cytotoxic T-cells
p59 ^{lmt}	Defective signalling in thymocytes but not peripheral T-cells
HOX 11	No spleen
IgE	No defects observed!
Fc ϵ R1 α -chain	Resistant to cutaneous and systemic anaphylaxis
IgM μ -chain membrane exon	Absence of B-cells
IL-6	No bone loss when ovariectomized (implications for osteoporosis?)
IL-18	Susceptible to <i>Leishmania major</i> ; shift from Th1 to Th2 response (decreased IFN γ and increased IL-4 production)
MHC class II A β	Decreased CD4 T-cells; inflammatory bowel disease
Perforin	Impaired CTL and NK cell function
TAP1	Lack CD8 cells
TNFR-1	Resistant to endotoxic shock; susceptible to <i>Listeria</i>
Modified from Brandon (1995) <i>Current Biology</i> 5, 625.	

example of a knocked in gene was seen on p. 133 with the replacement of endogenous RAG-2 with a RAG-2-GFP fusion gene.

Gene therapy in humans. We seem to be catching up with science fiction and are in the early stages of being able to correct genetic misfortune by the introduction of 'good' genes. Effective gene delivery still poses major problems but there have been some recent successes. One form of severe combined immunodeficiency (SCID) is due to a mutation in the γ c gene which encodes a subunit of the cytokine receptors for IL-2, -4, -7, -9 and -15. Correction of this defect in children has been achieved by *in vitro* transfer of the normal gene into CD34⁺ bone marrow stem cells using a vector derived from a Moloney retrovirus, a convincing proof of principle for human gene therapy. Adenovirus vectors have also proved popular in past attempts at gene therapy but are increasingly being replaced by adeno-associated virus (AAV) which may be safer. Some success has been achieved using intramuscular injection of AAV-based vectors to transfer human blood clotting factor IX into patients with hemophilia B.

SUMMARY

Isolation of leukocyte subpopulations

- Cells can be separated on the basis of physical characteristics such as size, buoyant density and adhesiveness.
- Phagocytic cells can be separated by a magnet after taking up iron particles, and cells which divide in response to a specific stimulus, e.g. antigen, can be eliminated by ultraviolet light after incorporation of 5-bromodeoxyuridine.
- Antibody-coated cells can be eliminated by complement-mediated cytotoxicity or anti-Ig-ricin conjugates; they can be isolated by panning on solid-phase anti-Ig or by cluster formation with magnetic beads bearing anti-Ig on their surface.
- Smaller numbers of cells can be fractionated by coating with a fluorescent monoclonal antibody and separating them from nonfluorescent cells in the FACS.
- Antigen-specific T-cells can be enriched as lines or clones by driving them with antigen; fusion to appropriate T-cell tumor lines yields immortal antigen-specific T-cell hybridomas.

Immunohistochemical localization of antigens in cells and tissues

- Antigens can be localized if stained by fluorescent antibodies and viewed in a fluorescence microscope.

- Confocal microscopy scans a very thin plane at high magnification and provides quantitative data on extremely sharp images of the antigen-containing structures which can also be examined in three dimensions.
- Antibodies can either be labeled directly or visualized by a second antibody, a labeled anti-Ig.
- In a flow cytometer single cells in individual droplets are interrogated by one or more lasers and quantitative data using different fluorescent labels can be logged, giving a complex phenotypic analysis of each cell in a heterogeneous mixture. In addition, forward scatter of the laser light defines cell size and 90° scatter, cell granularity.
- Fluorescent antibodies or their fragments can also be used for staining intracellular antigens in permeabilized cells. Intracellular probes for pH, Ca²⁺, Mg²⁺, Na⁺, thiols and DNA content are also available. Gene regulatory elements linked to a reporter gene such as GFP can also now be studied at the single-cell level by flow cytometry.
- Antibodies can be enzyme-labeled for histochemical definition of antigens at the light microscope level, and coupled with different-sized colloidal gold particles for ultrastructural visualization in the electron microscope.

(continued)

Gene expression

- mRNA expression can be localized by *in situ* hybridization using a complementary oligonucleotide probe.
- A complete picture of cellular gene expression is now attainable by hybridization to microarray chips.

Assessment of functional activity

- Neutrophil chemotaxis, phagocytosis, NADH (reduced nicotinamide adenine dinucleotide) oxidase activity and microbicidal potency can all be studied, almost on a routine basis.
- Lymphocyte responses to antigen are monitored by proliferation and/or cytokine release. Individual cells secreting cytokines can be identified by the ELISPOT technique in which the secreted product is captured by a solid-phase antibody and then stained with a second labeled antibody.
- Extracellular killing by cytotoxic T-cells, and NK cells, can be measured by the release of radioactive ^{51}Cr from prelabeled target cells.
- The precursor frequency of effector T-cells can be measured by staining the cells with peptide–MHC tetramers or by limiting dilution analysis.
- Antibody-forming cells can be enumerated, either by an immunofluorescence sandwich test or by plaque techniques in which the antibody secreted by the cells causes complement-mediated lysis of adjacent red cells, or is captured by solid-phase antigen in an ELISPOT assay.
- Functional activity can be assessed by cellular reconstitution experiments in which leukocyte sets and selected lymphoid tissue can be transplanted into unresponsive hosts such as X-irradiated recipients or SCID mice. Defined cell populations can also be separated and selectively recombined *in vitro*.
- Antibodies can be used to probe cellular function by cross-linking cell surface components or by selective destruction of particular intracellular sites by laser

irradiation of chromophore-conjugated specific antibodies which localize to the target area by penetrating permeabilized cells.

Genetic engineering of cells

- Genes can be inserted into mammalian cells by transfection using calcium phosphate precipitates, electroporation, liposomes and microinjection.
- Genes can also be taken into a cell after incorporation into vaccinia or retroviruses.
- Endogenous gene function can be inhibited by antisense RNA or by homologous recombination with a disrupted gene.
- Transgenic mice bearing an entirely new gene introduced into the fertilized egg by microinjection of DNA can be established as inbred lines.
- Genes can be introduced into embryonic stem cells; these modified stem cells are injected back into a blastocyst and can develop into founder mice from which pure transgenic animals can be bred. One very important application of this technique involves the disruption of a targeted gene in the embryonic stem cell by homologous recombination, producing 'knockout' mice lacking a specific gene. Conditional knockouts employ recombinase systems such as Cre/*loxP* in order to control the deletion either temporally or in a tissue-specific manner.
- 'Knock in' mice have a specified endogenous gene homologously replaced with either a variant of that gene or an entirely different gene.
- Human gene therapy promises an exciting future. Delivery of genes by vectors based on retroviruses or adeno-related virus is under intensive investigation.

See the accompanying website (www.roitf.com) for multiple choice questions.

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INTRODUCTION

In order to discuss the events which occur in the operation of the immune system as a whole, it is imperative to establish a nomenclature which identifies the surface markers on the cells involved since these are used for communication and are usually functional molecules reflecting the state of cellular differentiation. The nomenclature system is established as follows. Immunologists from the far corners of the world who have produced monoclonal antibodies directed to surface molecules on B- and T-cells, macrophages, neutrophils and natural killer (NK) cells, and so on, get together every so often to compare the specificities of their reagents in international workshops whose spirit of cooperation should be a lesson to most politi-

cians. Where a cluster of monoclonals are found to react with the same polypeptide, they clearly represent a series of reagents defining a given marker and are labeled with a CD (**cluster of differentiation**) number. Currently, there are nearly 250 CD numbers assigned, with some of them having subdivisions, but those in table 8.1 are most relevant to our discussions. A complete list of CD markers is given in Appendix 1. It is important to appreciate that the expression level of cell surface molecules often changes as cells differentiate or become activated and that 'subpopulations' of cells exist which differentially express particular molecules. When expressed at a low level the 'presence' or 'absence' of a given CD antigen may be rather subjective, but be aware that low level expression does not necessarily imply biological irrelevance.

THE NEED FOR ORGANIZED LYMPHOID TISSUE

For an effective immune response, an intricate series of cellular events must occur. Antigen must bind and if necessary be processed by antigen-presenting cells, which must then make contact with and activate T- and B-cells; T-helpers must assist B-cells and cytotoxic T-cell precursors, and there have to be mechanisms which amplify the numbers of potential effector cells by proliferation and then bring about differentiation to generate the mediators of humoral and cellular

immunity. In addition, memory cells for secondary responses must be formed and the whole response controlled so that it is adequate but not excessive and is appropriate to the type of infection being dealt with. By working hard, we can isolate component cells of the immune system and persuade them to carry out a number of responses to antigen in the test-tube, but compared with the efficacy of the overall development of immunity in the body, our efforts still leave much to be desired. *In vivo* the integration of the complex cellular interactions which form the basis of the immune response takes place within the organized architecture

Table 8.1. Some of the major clusters of differentiation (CD) markers on human cells.

CD	Expression	Functions
CD1	IDC, B subset	Presents glycolipid and other non-peptide antigens to T-cells
CD2	T, NK	Receptor for CD58 (LFA-3) costimulator. Binds sheep rbc
CD3	T	Transducing elements of T-cell receptor
CD4	T-helper, Mo, Mφ	MHC class II. HIV receptor
CD5	T, B subset	Involved in antigen receptor signaling
CD8	T-cytotoxic	MHC class I receptor
CD14	G, Mo, Mφ	LPS/LBP complex receptor
CD16	G, NK, B, Mφ, IDC	FcγRIII (medium affinity IgG receptor)
CD19	B, FDC	Part of B-cell antigen receptor complex
CD20	B	Unknown, but able to provide intracellular signals
CD21	B, FDC	CR2. Receptor for C3d and Epstein-Barr virus. Part of B-cell antigen receptor complex
CD23	B, Mo, FDC	FcεRII (low affinity IgE receptor)
CD25	*T, *B, *Mo, *Mφ	IL-2 receptor α chain
CD28	T, *B	Receptor for CD80/CD86 (B7.1 and B7.2) costimulators
CD32	Mo, Mφ, IDC, FDC, G, NK, B,	FcγRII (low affinity IgG receptor)
CD34	Progenitors	Adhesion molecule. Stem cell marker
CD40	B, Mφ, IDC, FDC	Receptor for CD40L costimulator
CD45RA	Resting/Naive T-cells, B, G, Mo, NK	Phosphatase, cell activation
CD45RO	Activated/Memory T-cells, Mo, DC	Phosphatase, cell activation
CD64	Mo, Mφ, DC	FcγRI (high affinity IgG receptor)
CD79a/CD79b	B	Transducing elements of B-cell receptor
CD80	*B, *T, Mφ, DC	B7.1 receptor for CD28 costimulator and for CTLA4 inhibitory signal
CD86	B, IDC, Mo	B7.2 receptor for CD28 costimulator and for CTLA4 inhibitory signal
CD95	Widespread	Fas receptor for FasL. Transmits apoptotic signals

*, activated; B, B-lymphocytes; FDC, follicular dendritic cells; G, granulocytes; IDC, interdigitating dendritic cells; Mast, mast cells; Mφ, macrophages; Mo, monocytes; NK, natural killer cells; T, T-lymphocytes.

of peripheral, or secondary, lymphoid tissue which includes the lymph nodes, spleen and unencapsulated tissue lining the respiratory, gastrointestinal and genitourinary tracts.

These tissues become populated by cells of reticular origin and by macrophages and lymphocytes derived

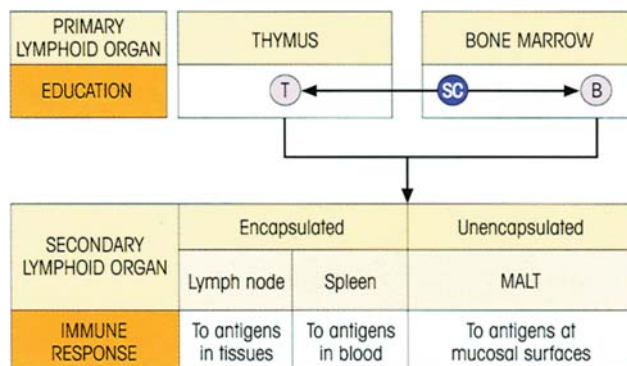


Figure 8.1. The functional organization of lymphoid tissue. Stem cells (SC) arising in the bone marrow differentiate into immunocompetent T- and B-cells in the primary lymphoid organs and then colonize the secondary lymphoid tissues where immune responses are organized. The mucosal-associated lymphoid tissue (MALT) produces antibodies for mucosal secretions.

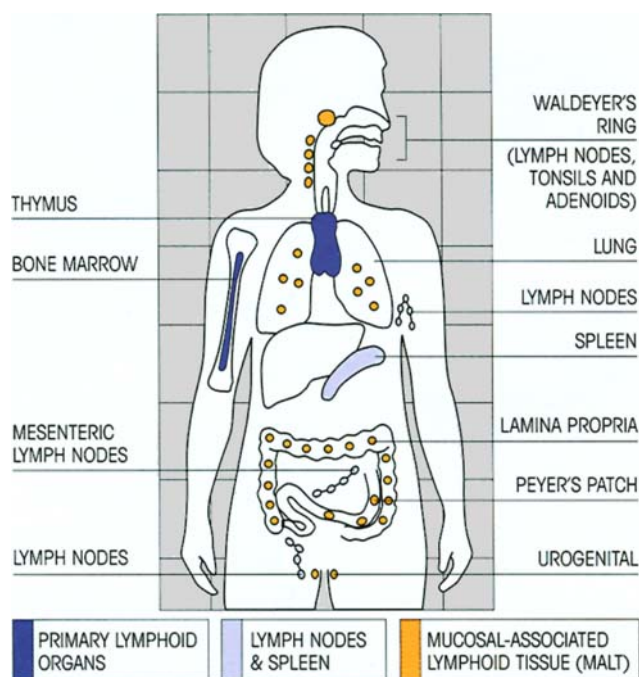


Figure 8.2. The distribution of major lymphoid organs and tissues throughout the body.

from bone marrow stem cells, the T-cells first differentiating into immunocompetent cells by a high-pressure training period in the thymus, the B-cells undergoing their education in the bone marrow itself (figure 8.1). In essence, the lymph nodes filter off and, if necessary, respond to foreign material draining body tissues, the spleen monitors the blood and the unencapsulated lymphoid tissue is strategically integrated into mucosal surfaces of the body as a forward defensive system based on IgA secretion.

The anatomical disposition of these lymphoid tissues is illustrated in figure 8.2. The lymphatics and associated lymph nodes form an impressive network, draining the viscera and the more superficial body structures before returning to the blood by way of the thoracic duct (figure 8.3).

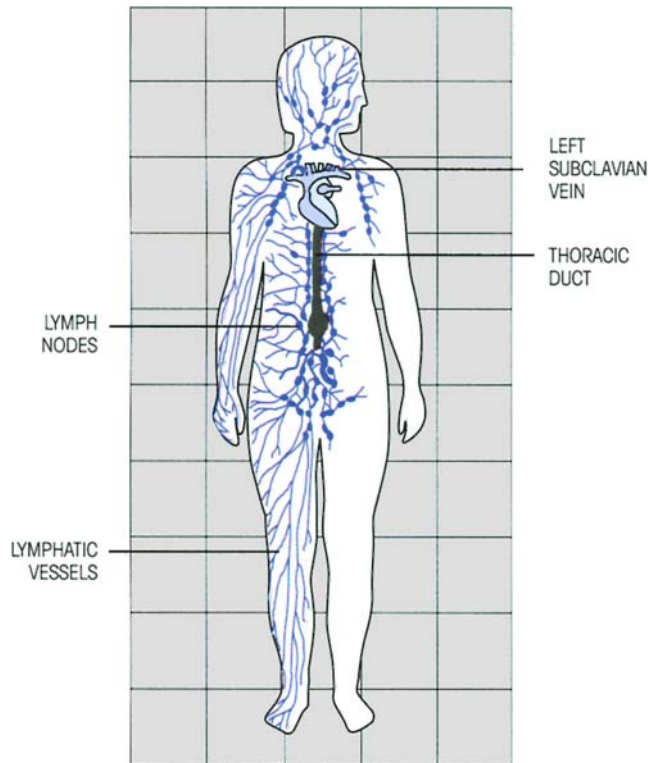


Figure 8.3. The network of lymph nodes and lymphatics. Lymph nodes occur at junctions of the draining lymphatics. The lymph finally collects in the thoracic duct and thence returns to the bloodstream via the left subclavian vein.

Communication between these tissues and the rest of the body is maintained by a pool of recirculating lymphocytes which pass from the blood into the lymph nodes, spleen and other tissues and back to the blood by the major lymphatic channels such as the thoracic duct (figures 8.4 and 8.10).

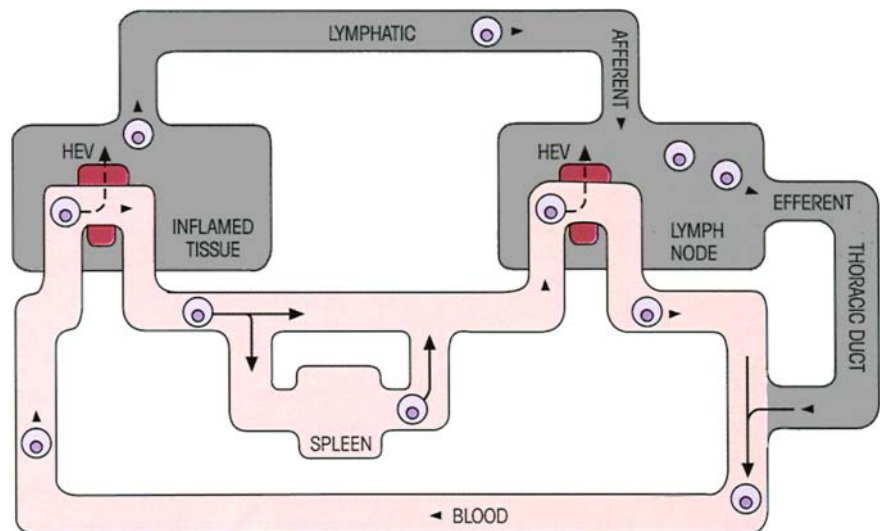
LYMPHOCYTES TRAFFIC BETWEEN LYMPHOID TISSUES

This traffic of lymphocytes between the tissues, the bloodstream and the lymph nodes enables antigen-sensitive cells to seek the antigen and to be recruited to sites at which a response is occurring, while the dissemination of memory cells and their progeny enables a more widespread response to be organized throughout the lymphoid system. Thus, antigen-reactive cells are depleted from the circulating pool of lymphocytes within 24 hours of antigen first localizing in the lymph nodes or spleen; several days later, after proliferation at the site of antigen localization, a peak of activated cells appears in the thoracic duct. When antigen reaches a lymph node in a primed animal, there is a dramatic fall in the output of cells in the efferent lymphatics, a phenomenon described variously as 'cell shutdown' or 'lymphocyte trapping' and which is thought to result from the antigen-induced release of soluble factors from T-cells (cf. the cytokines, p. 179); this is followed by an output of activated blast cells which peaks at around 80 hours.

Lymphocytes home to their specific tissues

Naive lymphocytes enter a lymph node through the afferent lymphatics and by guided passage across the

Figure 8.4. Traffic and recirculation of lymphocytes through encapsulated lymphoid tissue and sites of inflammation. Blood-borne lymphocytes enter the tissues and lymph nodes passing through the high-walled endothelium of the postcapillary venules (HEV) and leave via the draining lymphatics. The efferent lymphatics, finally emerging from the last node in each chain, join to form the thoracic duct which returns the lymphocytes to the bloodstream. In the spleen, which lacks HEVs, lymphocytes enter the lymphoid area (white pulp) from the arterioles, pass to the sinusoids of the erythroid area (red pulp) and leave by the splenic vein. Traffic through the mucosal immune system is elaborated in figure 8.10.



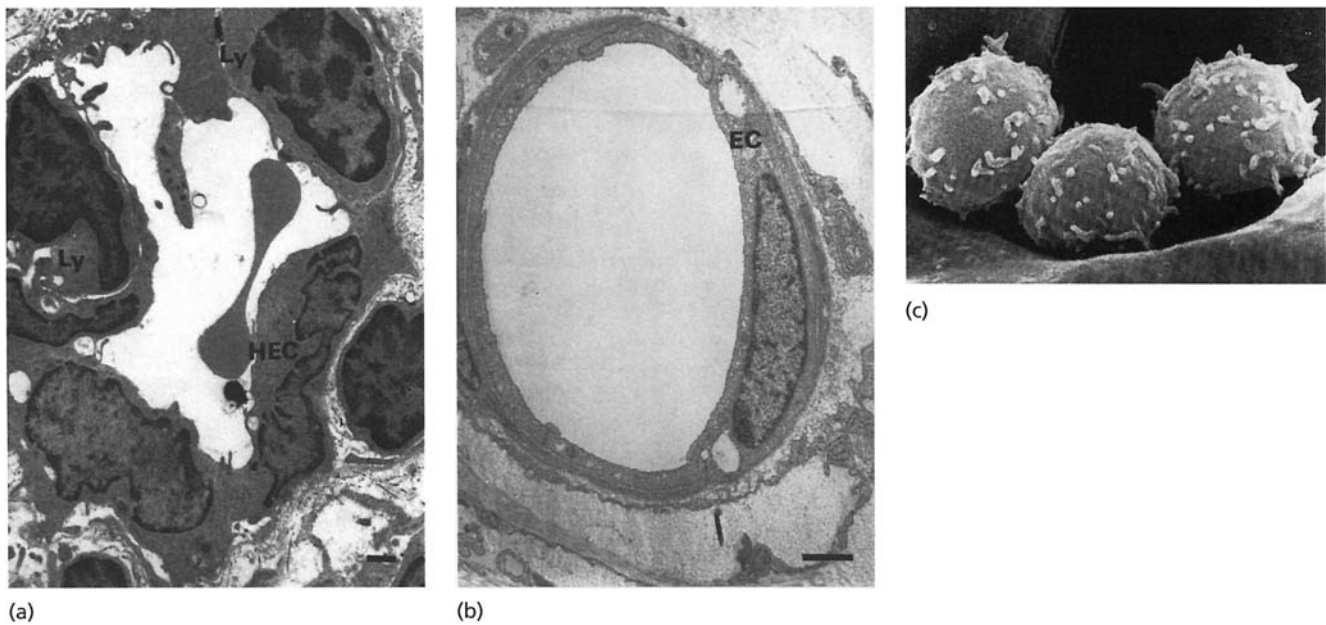


Figure 8.5. Lymphocyte association with postcapillary venules. (a) High-walled endothelial cells (HEC) of postcapillary venules in rat cervical lymph nodes showing intimate association with lymphocytes (Ly). (b) Flattened capillary endothelial cell (EC) for comparison. (c) Lymphocytes adhering to HEC (scanning electron micrograph). ((a) and (b) Kindly provided by Dr Ann Ager and (c) by Dr W. van Ewijk.)

specialized **high-walled endothelium of the postcapillary venules (HEVs)** (figure 8.5). Comparable HEVs offer the transit of cells concerned in mucosal immunity to Peyer's patches. In other cases involving migration into normal and inflamed tissues, the lymphocytes bind to and cross nonspecialized flatter endothelia. Lymphoblasts and memory cell populations display tissue-restricted migration to extra-lymphoid sites such as skin or mucosal epithelium, while lymphocytes, as well as neutrophils and monocytes, target and migrate into sites of inflammation in response to locally produced mediators.

This highly organized traffic is orchestrated by directing the relevant lymphocytes to different parts of the lymphoid system and the various other tissues by a series of **homing receptors** which include members of the **integrin** superfamily (table 8.2) and also a member of the selectin family, L-selectin. Integrins can bind to extracellular matrix, plasma proteins and to other cell surface molecules, and they are widely involved in embryogenesis, cell growth, differentiation, adhesion, motility, programmed cell death and tissue maintenance. Within the immune system their complementary ligands include cell surface **vascular addressins** present on the appropriate blood vessel endothelium (figure 8.6). These act as selective gateways which allow partic-

ular populations of lymphocytes access to the appropriate tissue (table 8.3). Chemokines such as SLC (secondary lymphoid tissue chemokine), presented by vascular endothelium, play a key role in triggering lymphocyte arrest, the chemokine receptors on the lymphocyte being involved in the functional upregulation of integrins. With respect to lymphocytes and dendritic cells which are destined for the skin, E-selectin on the endothelium recognizes cutaneous leukocyte antigen (CLA) on these cells.

Transmigration occurs in three stages

Step 1: Tethering and rolling

In order for the lymphocyte to become attached to the endothelial cell, it has to overcome the shear forces created by the blood flow. This is effected by a force of attraction between the homing receptors and their ligands on the vessel wall which operates through microvilli on the leukocyte surface (figure 8.6). After this tethering process, the lymphocyte rolls along the endothelial cell, with the integrins VLA-4 or LPAM-1 on the lymphocyte binding to their ligands on the endothelium.

Step 2: LFA-1 activation and cell flattening

This process leads to activation and recruitment of LFA-1 to the nonvillous surface of the lymphocyte. This integrin binds very strongly to ICAM-1 and -2 on the endothelial cell, the intimate contact causing the lymphocyte to flatten.

Table 8.2. The integrin superfamily. In general, the integrins are concerned with intercellular adhesion and adhesion to extracellular matrix components. Many of them are also involved in cell signal transduction. They are heterodimers with unique but related α chains which can be grouped into subsets, each of which has a common β chain. The VLA subfamily took its name from VLA-1 and -2 which appeared as very late antigens (VLA) on T-cells, 2–4 weeks after *in vitro* activation. However, VLA-3, -4 and -5 belong to the same family but are not ‘very late’ and are found to different extents on lymphocytes, monocytes, platelets and probably hematopoietic progenitors. A structure called the I (inserted) domain is present in many integrin subunits and contains the *metal ion-dependent adhesion site* (MIDAS) which, in the presence of Mg^{2+} , is involved in binding the Arg.Gly.Asp. (RGD) motif on many of the ligands essential for cell adhesion.

Integrin	Subunits	*CD group	Expression	Ligand
VLA proteins				
VLA-1	$\alpha_1\beta_1$	CD49a/CD29	Widespread	LM, CO
VLA-2	$\alpha_2\beta_1$	CD49b/CD29	Widespread	LM, CO
VLA-3	$\alpha_3\beta_1$	CD49c/CD29	Widespread	FN, LM, CO, EN
VLA-4	$\alpha_4\beta_1$	CD49d/CD29	Widespread	FN, VCAM-1
VLA-5	$\alpha_5\beta_1$	CD49e/CD29	Widespread	FN
VLA-6	$\alpha_6\beta_1$	CD49f/CD29	Widespread	LM
Leukocyte integrins				
LFA-1	$\alpha_L\beta_2$	CD11a/CD18	Leukocytes	ICAM-1, -2, -3
CR3 (Mac-1)	$\alpha_M\beta_2$	CD11b/CD18	N, Mo, LGL	ICAM-1, C3bi, FG, FX
p150,95	$\alpha_X\beta_2$	CD11c/CD18	M, Mo, N	C3bi, LPS
	$\alpha_D\beta_2$	CD11d/CD18	Tissue M	ICAM-3
LPAM-1	$\alpha_4\beta_7$	CD49d/ND	Lymphocytes	MAcCAM-1, VCAM-1, FN
	$\alpha_E\beta_7$	CD103/ND	Intraepithelial T-cells	E-cadherin
Cytoadhesins				
GPIIb/IIIa	$\alpha_{IIb}\beta_3$	CD41/CD61	Megakaryocytes, platelets	FN, VN, FG, VWF, THR
VN receptor	$\alpha_v\beta_3$	CD51/CD61	Widespread	VN, FN, FG, VWF, THR, TN

CO, collagen; CR3, complement receptor 3; EN, entactin; FG, fibrinogen; FN, fibronectin; FX, factor X; GPIIb/IIIa, integrin glycoproteins IIb and IIIa; ICAM, intercellular adhesion molecule; LFA, lymphocyte function-associated molecule; LGL, large granular lymphocyte; LM, laminin; LPAM, lymphocyte Peyer’s patch adhesion molecule; M, macrophage; MAcCAM, mucosal addressin cell adhesion molecule; Mo, monocyte; N, neutrophil; THR, thrombospondin; TN, tenascin; VCAM, vascular cell adhesion molecule; VLA, very late antigen (although they are not all expressed late!); VN, vitronectin; VWF, von Willebrand factor. *CD markers are explained on p. 147. ND, no CD designation yet assigned.

Step 3: Diapedesis

The flattened lymphocyte now uses the LFA-1–ICAM interaction and the immunoglobulin superfamily member PECAM-1 (platelet endothelial cell adhesion molecule, CD31; not only present on platelets, see Appendix 1), to elbow its way between the endothelial cells and into the tissue in response to a chemotactic signal.

A closer look at the interacting receptors and their ligands

The ligands for selectins tend to be mucinous molecules covered with dense patches of O-linked sugars. The selectins generally terminate in a lectin domain (hence ‘selectin’), as might be expected given the nature of the ligands.

P- and E-selectins allow memory T-cells expressing high levels of VLA-4 to localize at sites of inflammation where subsequent interaction with the upregulated vascular cell adhesion molecule (VCAM-1) on the inflamed endothelium, and with various cytokine and chemokine inflammatory mediators presented on the surface of extracellular matrix proteoglycans, leads to lymphocyte transmigration. Secretion of heparanase by the stimulated T-cell could break down elements of the extracellular matrix, thereby facilitating movement towards the sounds of inflammatory gunfire.

Homing previously activated and memory lymphocytes to sites of inflammation provoked by infectious agents makes a great deal of sense. The same may be said for the mechanisms which enable lymphocytes bearing receptors for mucosal-associated lymphoid tissue (MALT) to circulate within and between the collections of lymphoid tissue guarding the external body surfaces (see figure 8.10). In this way, lymphocytes, such as those programed to support the synthesis of IgA destined for secretion, will not waste time cooling their heels in encapsulated peripheral lymph nodes which play no role in mucosal protection.

ENCAPSULATED LYMPH NODES

The encapsulated tissue of the lymph node contains a meshwork of reticular cells and their fibers organized into sinuses. These act as a filter for lymph draining the body tissues, and possibly bearing foreign antigens, which enters the subcapsular sinus by the afferent vessels and diffuses past the lymphocytes in the cortex to reach the macrophages of the medullary sinuses (figure 8.7) and thence the efferent lymphatics (figures 8.4 and 8.7). What is so striking about the organization of the lymph node is that the T- and B-lymphocytes are very largely separated into different anatomical compartments.

B-cell areas

The follicular aggregations of B-lymphocytes are a prominent feature of the outer cortex. In the unstimulated node they are present as spherical collections of cells termed **primary follicles** (figure 8.7i), but after

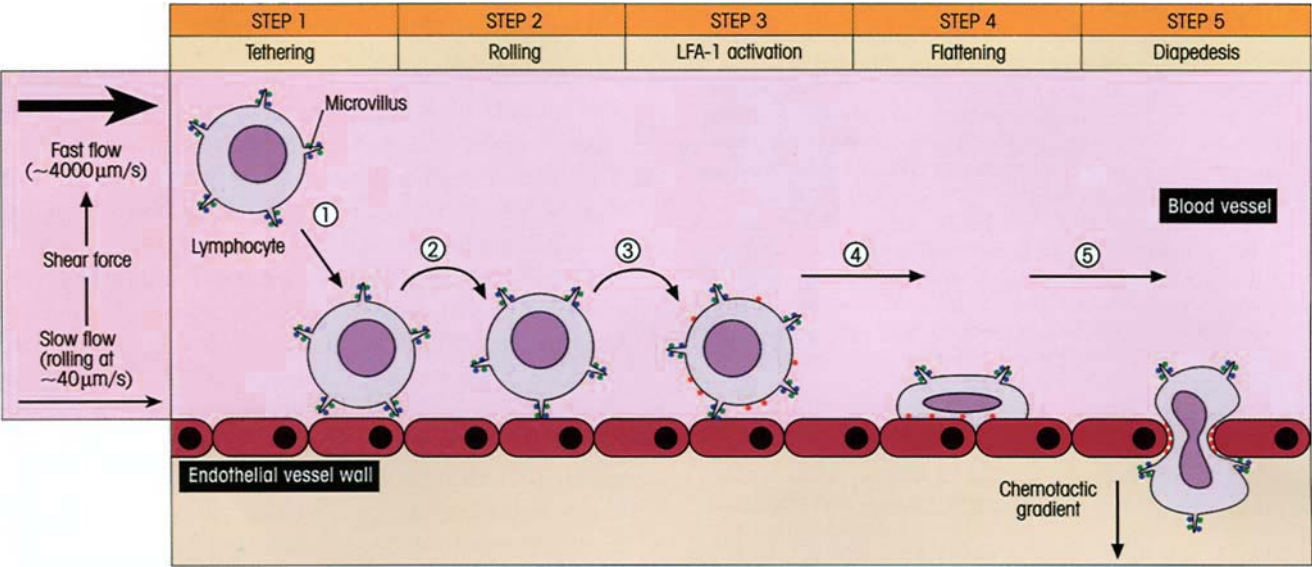


Figure 8.6. Homing and transmigration of lymphocytes. Fast-moving lymphocytes are tethered (Step 1) to the vessel walls of the tissue they are being guided to enter through an interaction between specific homing receptors, such as L-selectin (•) or LPAM-1 (•) located on the microvilli of the lymphocyte, and their ligands, for example GlyCAM-1 or MAdCAM-1, on the endothelium of the vessel wall. After rolling along the surface of the endothelial cells (Step 2),

activation of the lymphocyte LFA-1 integrin (cf. table 8.2) occurs (Step 3, •). Note that, because this integrin is absent from the microvilli, firm binding occurs by the body of the lymphocyte to its ligands, ICAM-1/2, on the endothelium. This process results in cell flattening (Step 4) and migration of the lymphocyte between adjacent endothelial cells, a process referred to as diapedesis (Step 5).

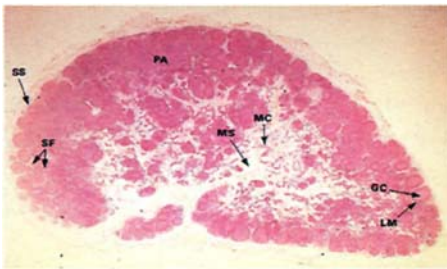
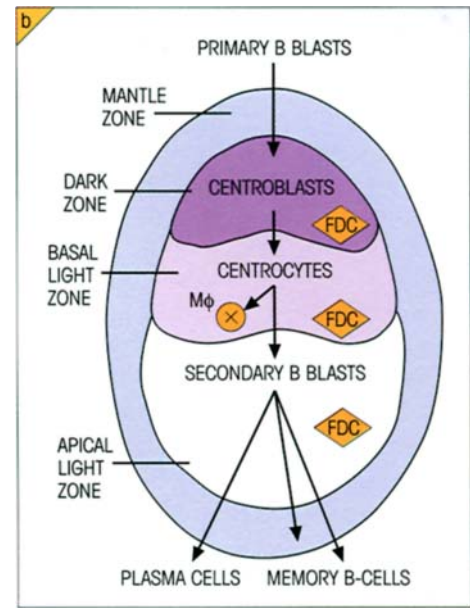
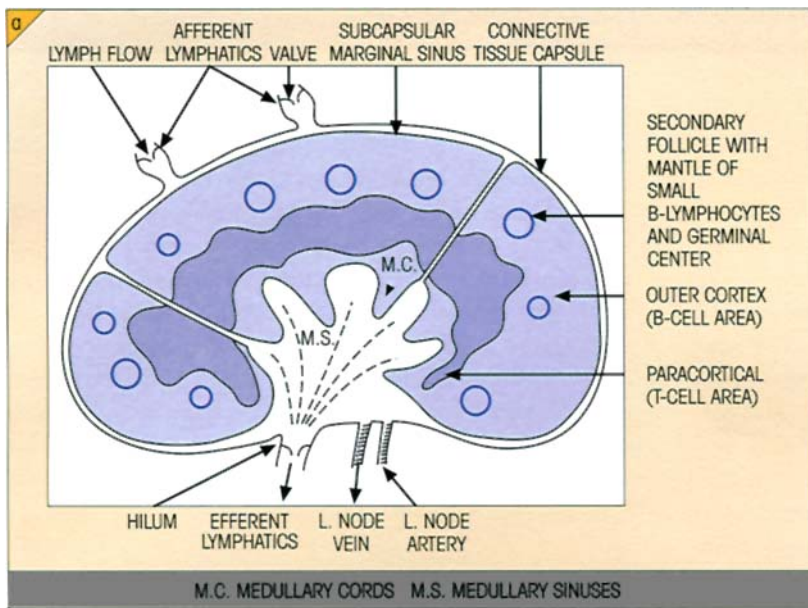
Table 8.3. Ligand–receptor interactions determining the homing of defined lymphocyte populations.

Homing	*Ligand on endothelium	Receptor on lymphocyte	Result of interaction				
			Tethering	Rolling	LFA-1 activation	Flattening	Diapedesis
Naive and memory T-cells into a lymph node	GlyCAM-1 VCAM-1 SLC ICAM-1/2	L-selectin VLA-4 CCR7 LFA-1					
Naive T-cells into Peyer's patches	MAdCAM-1 MAdCAM-1 SLC ICAM-1/2	L-selectin LPAM-1 CCR7 LFA-1					
LPAM-1 ^{hi} lymphoblasts and memory T-cells into Peyer's patch and lamina propria	MAdCAM-1 ICAM-1/2	LPAM-1 LFA-1					
LPAM-1 ^{hi} memory T-cells into site of inflammation	P-selectin E-selectin VCAM-1 ICAM-1/2	PSGL-1 and ESL-1 ESL-1 VLA-4 LFA-1					

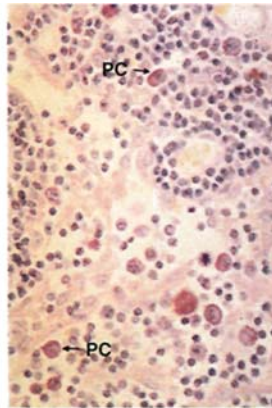
*In lymph node and Peyer's patches the molecules are present on the specialized HEV. CCR7, CC subgroup chemokine receptor-7; ESL, E-selectin ligand; GlyCAM, glycosylation-dependent cell adhesion molecule; PSGL, P-selectin glycoprotein ligand; SLC, secondary lymphoid tissue chemokine.

antigenic challenge they form **secondary follicles** (figure 8.7f) which consist of a corona or mantle of concentrically packed, resting, small B-lymphocytes possessing both IgM and IgD on their surface surrounding a pale-staining **germinal center** (figure 8.7b). This contains large, usually proliferating, B-blasts, a minority

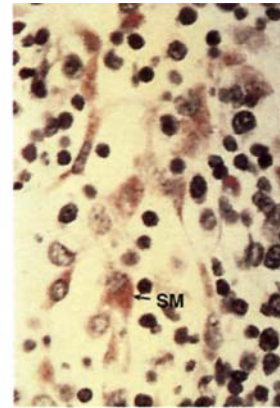
of T-cells, scattered conventional reticular macrophages containing 'tingible bodies' of phagocytosed lymphocytes, and a tight network of specialized **follicular dendritic cells** (FDCs) with elongated cytoplasmic processes and few, if any, lysosomes. Germinal centers are greatly enlarged in secondary antibody



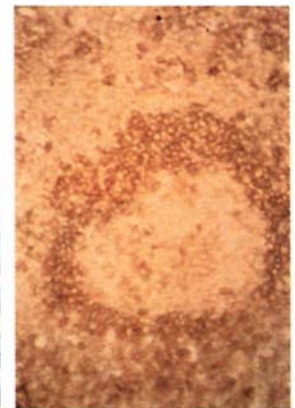
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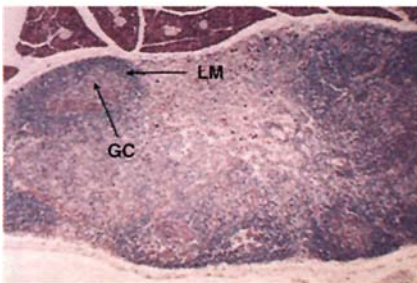
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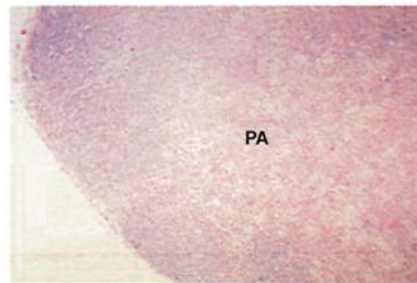
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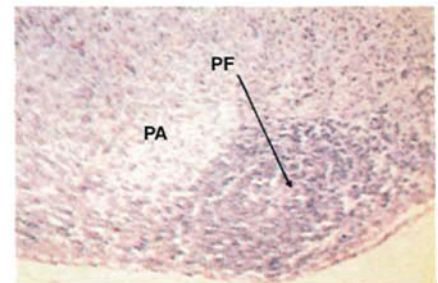
(f)



(g)



(h)



(i)

Figure 8.7. Lymph node. (a) Diagrammatic representation of section through a whole node. (b) Diagram showing differentiation of B-cells during passage through different regions of an active germinal center. FDC, follicular dendritic cell; Mφ, macrophage; ×, apoptotic B-cell. (c) Human lymph node, low-power view. (d) Medulla stained with methyl green (DNA)/pyronin (RNA) to show the basophilic (pink) cytoplasm of the plasma cells with their abundant ribosomes. (e) Medullary sinus of lymph node draining site of lithium carmine injection showing macrophages which have phagocytosed the colloidal dye (one is arrowed). (f) Secondary lymphoid follicle showing germinal center surrounded by a mantle of small B-lymphocytes stained by anti-human IgD labeled with horseradish peroxidase (brown color). There are few IgD-positive cells in the center but both areas contain IgM-positive B-lymphocytes. (g) Node from mouse immunized with the thymus-independent antigen, *Pneumococcus* polysaccharide SIII, revealing prominent stimulation of secondary follicles with germinal centers. (h) Methyl green/pyronin stain of lymph node draining site of skin painted with the contact sensitizer oxazolone, highlighting the generalized expansion and activation of the paracortical T-cells, the T-blasts being strongly basophilic. (i) The same study in a neonatally thymectomized mouse shows a lonely primary nodule (follicle) with complete lack of cellular response in the paracortical area. GC, germinal center; LM, lymphocyte mantle of secondary follicle; MC, medullary cords; MS, medullary sinus; PA, paracortical area; PC, plasma cell; PF, primary follicle; SF, secondary follicle; SM, sinusoidal macrophage; SS, subcapsular sinus. ((c) Photographed by Professor P.M. Lydyard; (d–f) by Dr K.A. MacLennan; (g–i) courtesy of Dr M. de Sousa and Professor D.M.V. Parrott.)

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responses during which they constitute sites of B-cell maturation and the generation of B-cell memory.

In the absence of antigen drive, the primary follicles are composed of a mesh of FDCs whose spaces are filled with recirculating, but resting, small B-lymphocytes. On priming with a single dose of a T-dependent antigen (i.e. antigen for which the B-cells require cooperation from T-helper cells; cf. p. 171), the FDC network can be colonized by as few as three primary B-blasts which undergo exponential growth, producing around 10^4 so-called centroblasts and displacing the original resting B-cells which now form the follicular mantle. These highly mitotic centroblasts, with no surface IgD (sIgD) and very little sIgM, then differentiate into light zone centrocytes which are non-cycling and begin to upregulate their expression of sIg. At this stage there is very extensive apoptotic cell death, giving rise to DNA fragments which are visible as 'tingible bodies' within the macrophages, the final resting place of the dead cells. The survivors undergo their final training in the apical light zone. A proportion of those which are shunted down the **memory** cell pathway take up residence in the mantle zone population, the remainder joining the recirculating B-cell pool. Other cells differentiate into plasmablasts with a well-defined endoplasmic reticulum, prominent Golgi apparatus and cytoplasmic Ig; these migrate to become plasma cells in the medullary cords which project between the medullary sinuses (figure 8.7d). This maturation of antibody-forming cells at a site distant from that at which antigen triggering has occurred is also seen in the spleen, where plasma cells are found predominantly in the marginal zone. One's guess is that this movement of cells acts to prevent the generation of high local concentrations of antibody within the germinal center, so avoiding neutralization of the antigen on the FDCs and premature shutting off of the immune response.

The remainder of the outer cortex is also essentially a B-cell area with scattered T-cells.

T-cell areas

T-cells are mainly confined to a region referred to as the paracortical (or thymus-dependent) area (figure 8.7a); in nodes taken from children with selective T-cell deficiency (figure 15.5), or from neonatally thymectomized mice (figure 8.7i), the paracortical region is seen to be virtually devoid of lymphocytes. Furthermore, when a T-cell-mediated response is elicited in a normal animal, say by a skin graft or by painting chemicals such as picryl chloride on the skin to induce contact hypersensitivity, there is a marked prolifera-

tion of cells in the thymus-dependent area and typical lymphoblasts are evident (figure 8.7h). In contrast, stimulation of antibody formation by the thymus-independent antigen, *Pneumococcus* polysaccharide SIII, leads to proliferation in the cortical lymphoid follicles with the development of germinal centers, while the paracortical region remains inactive, reflecting the inability to develop cellular hypersensitivity to the polysaccharide (figure 8.7g). As expected, nodes taken from children with congenital hypogammaglobulinemia associated with failure of B-cell development conspicuously lack primary and secondary follicles.

SPLEEN

On a fresh section of spleen, the lymphoid tissue forming the white pulp is seen as circular or elongated gray areas (figure 8.8b) within the erythrocyte-filled red pulp which consists of splenic cords lined with macrophages and venous sinusoids. As in the lymph node, T- and B-cell areas are segregated (figure 8.8a). The spleen is a very effective blood filter removing effete red and white cells and responding actively to blood-borne antigens, the more so if they are particulate. Plasmablasts and mature plasma cells are present in the marginal zone extending into the red pulp (figure 8.8c).

MUCOSAL-ASSOCIATED LYMPHOID TISSUE (MALT)

The respiratory, gastrointestinal and genitourinary tracts are guarded immunologically by sub-epithelial accumulations of lymphoid tissue which are not constrained by a connective tissue capsule (figure 8.9). These may occur as diffuse collections of lymphocytes, plasma cells and phagocytes throughout the lung and the lamina propria of the intestinal wall (figure 8.9a and b), or as more clearly organized tissue with well-formed follicles. In humans, the latter includes the lingual, palatine and pharyngeal tonsils (figure 8.9c), the Peyer's patches of the small intestine (figure 8.9d) and the appendix. MALT forms an interconnected secretory system within which cells committed to IgA or IgE synthesis may circulate (figure 8.10).

In the gut, antigen enters the Peyer's patches (figure 8.9d) across specialized epithelial cells (cf. figure 8.15) and stimulates the antigen-sensitive lymphocytes. After activation these drain into the lymph and, after a journey through the mesenteric lymph nodes and the thoracic duct, they pass from the bloodstream into the

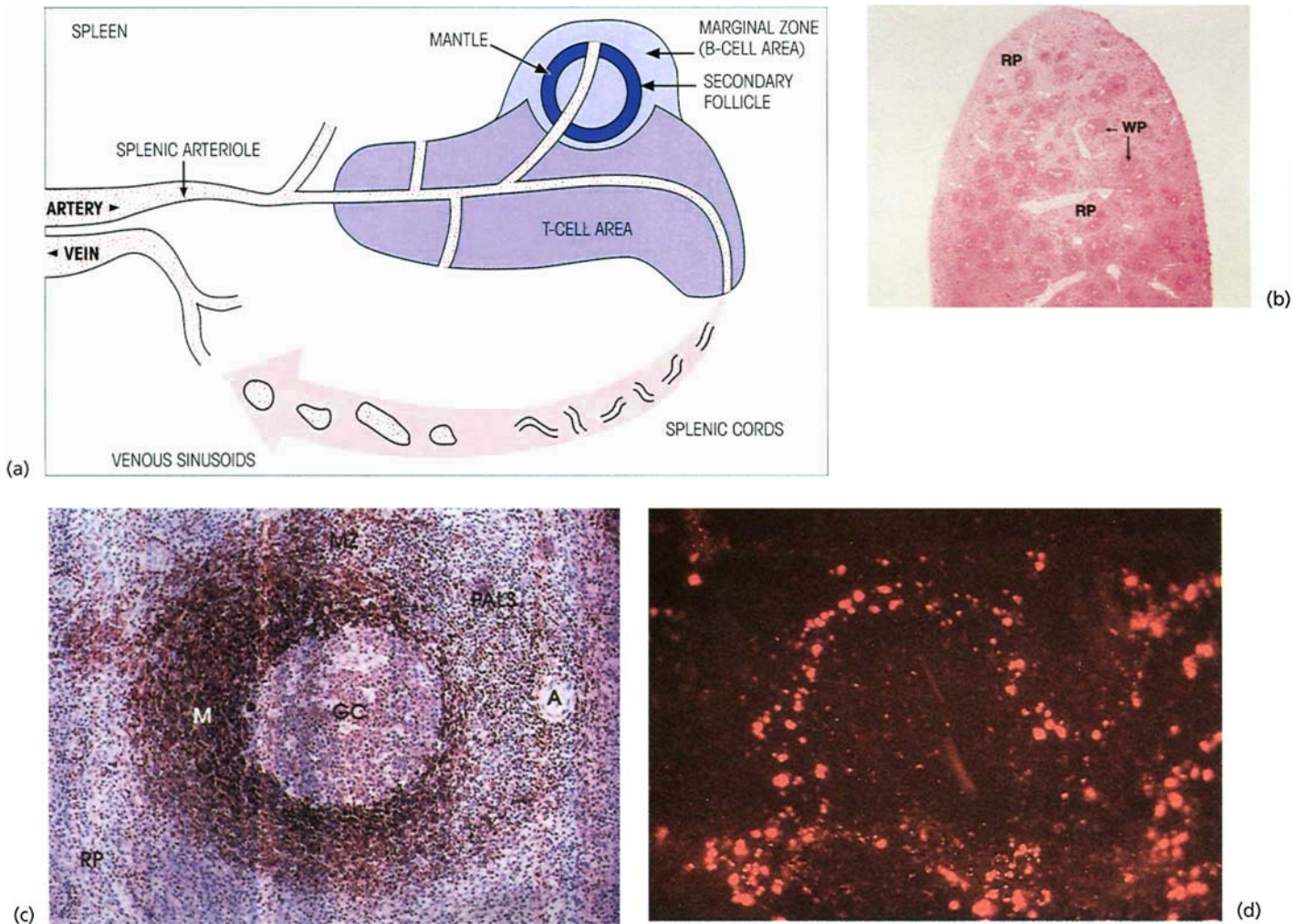


Figure 8.8. Spleen. (a) Diagrammatic representation. (b) Low-power view showing lymphoid white pulp (WP) and red pulp (RP). (c) High-power view of germinal center (GC) and lymphocyte mantle (M) surrounded by marginal zone (MZ) and red pulp (RP). Adjacent to the follicle, an arteriole (A) is surrounded by the periarteriolar lymphoid sheath (PALS) predominantly consisting of T-

cells. Note that the marginal zone is only present above the secondary follicle. (d) Localization of the thymus-independent antigen, ficoll, on the marginal zone macrophages. The ficoll is visualized by labeling with the red fluorescent dye tetramethyl-rhodamine. ((b) Photographed by Professor P.M. Lydyard; (c) by Professor I.C.M. MacLennan; (d) kindly provided by Professor J.H. Humphrey.)

lamina propria (figure 8.10) where they become IgA-forming cells which, because they are now broadly distributed, protect a wide area of the bowel with protective antibody. The cells also appear in the lymphoid tissue of the lung and in other mucosal sites guided by the interactions of specific homing receptors with appropriate HEV addressins as discussed earlier. Similarly, intranasal immunization is particularly effective at generating antibody production in the genitourinary tract.

Intestinal lymphocytes

The intestinal **lamina propria** is home to a predominantly activated T-cell population rich in the LPAM-1

integrin (table 8.2), the ligand for MAdCAM-1 on the lamina propria postcapillary venules (figure 8.11). These T-cells bear a phenotype roughly comparable to that of peripheral blood lymphocytes: viz. > 95% T-cell receptor (TCR) $\alpha\beta$ and a CD4:CD8 ratio of 7:3. Within the lamina propria there is also a generous sprinkling of activated B-blasts and plasma cells secreting IgA for transport by the poly-Ig receptor to the intestinal lumen (cf. p. 53).

Intestinal **intraepithelial lymphocytes** (IELs) are quite a different kettle of fish. They are also mostly T-cells, 10–40% of which have a $\gamma\delta$ TCR. Of those bearing an $\alpha\beta$ TCR, most are CD8 positive and can be divided into two populations. One-third of them possess the conventional form of CD8, which is a heterodimer

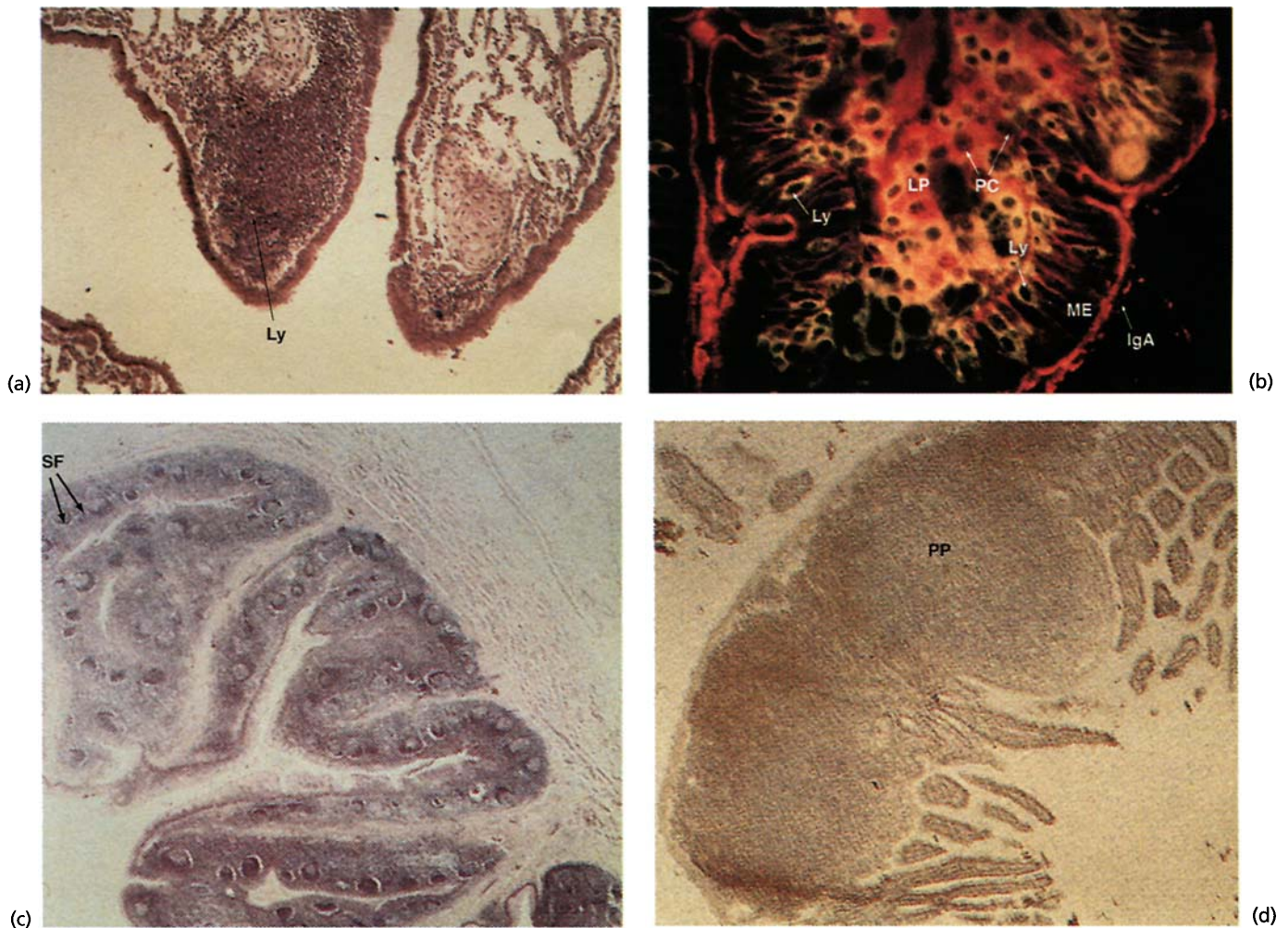


Figure 8.9. The IgA secretory immune system (MALT). (a) Section of lung showing a diffuse accumulation of lymphocytes (Ly) in the bronchial wall. (b) Section of human jejunum showing lymphoid cells (Ly), stained green by a fluorescent antileukocyte monoclonal antibody, in the mucosal epithelium (ME) and in the lamina propria (LP). A red fluorescent anti-IgA conjugate stains the cytoplasm of plasma cells (PC) in the lamina propria and detects IgA in the surface

mucus; altogether a super picture! (c) Low-power view of human tonsil showing the MALT with numerous secondary follicles (SF) containing germinal centers. (d) Peyer's patches (PP) in mouse ileum. The T-cell areas are stained brown by a peroxidase-labeled monoclonal antibody to Thy 1. ((a) Kindly provided by Professor P. Lydyard; (b) by Professor G. Jannosy; (c) by Mr C. Symes; and (d) by Dr E. Andrew.)

composed of a CD8 α chain and a CD8 β chain. However, two-thirds of them instead express a CD8 $\alpha\alpha$ homodimer which is almost exclusively found only on IELs. Whilst the CD8 $\alpha\beta$ IELs are restricted by classical MHC class I molecules, the CD8 $\alpha\alpha$ IELs appear to recognize nonclassical MHC molecules (cf. p. 77) perhaps including TL and Qa1 but not, apparently, CD1. The antigen specificity of most IELs is unknown. Intraepithelial lymphocytes and intraepithelial dendritic cells express high levels of the $\alpha\beta$ integrin which binds E-cadherin on intestinal epithelial cells.

The relatively high proportion of TCR $\gamma\delta$ cells is also unusual and most of these also express the CD8 $\alpha\alpha$ characteristic of IELs. Since a number of cloned $\gamma\delta$ T-

cells have been found to have specificity for heat-shock proteins, which are widely distributed in nature and usually highly immunogenic, it has been postulated that they act as a relatively primitive first line of defense at the outer surfaces of the body.

Reflect for a moment on the fact that roughly 10^{14} bacteria reside in the intestinal lumen of the normal adult human. That is a pretty impressive number of 'noughts' to swallow. Yet combined with the barrier of mucins produced by goblet cells and the protective zone of secreted IgA antibodies, these collections of intestinal lymphocytes represent a crucial line of defense.

It is not only the intestine that has a localized

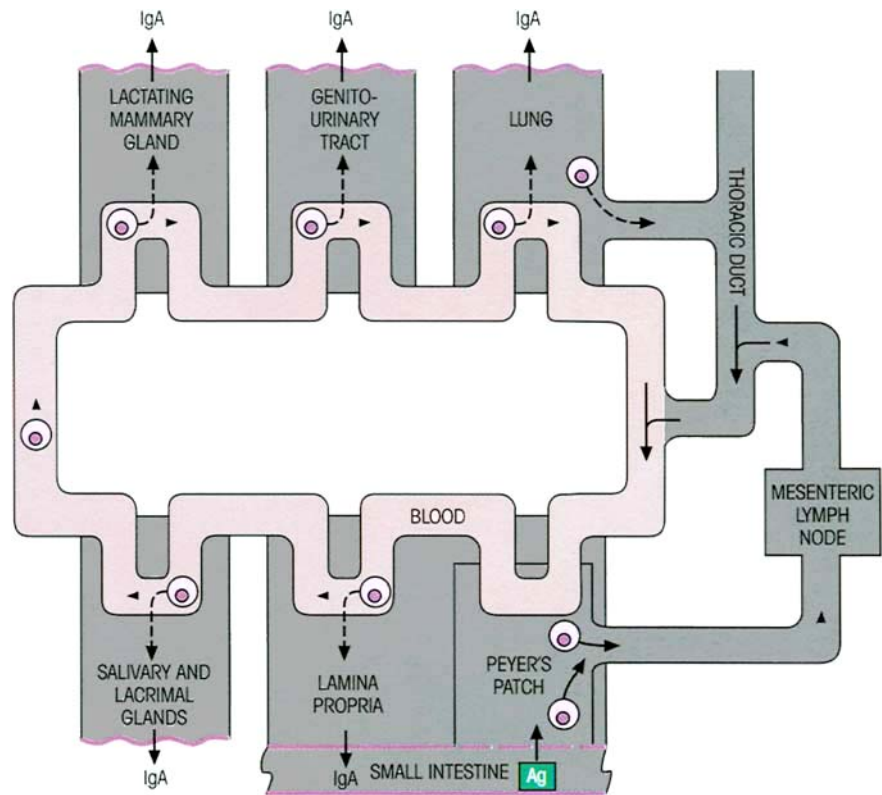
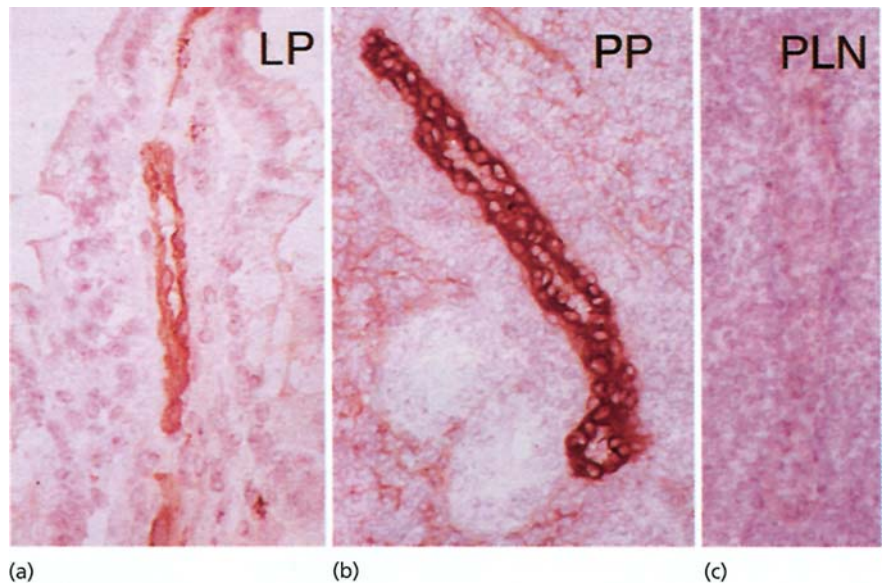


Figure 8.10. Circulation of lymphocytes within the mucosal-associated lymphoid system. Antigen-stimulated cells move from Peyer's patches to colonize the lamina propria and the other mucosal surfaces (), forming what has been described as a common mucosal immune system.

Figure 8.11. Selective expression of the mucosal vascular addressin MAdCAM-1 on endothelium involved in lymphocyte homing to gastrointestinal sites. Immunohistologic staining reveals the presence of MAdCAM-1 (a) on postcapillary venules in the small intestinal lamina propria and (b) on HEV in Peyer's patches, but its absence from (c) HEV in peripheral lymph nodes. (Reproduced with permission from Butcher E.C. *et al.* (1999) *Advances in Immunology* 72, 209.) At least some component of intestinal trafficking appears to operate as a subcomponent of a common mucosal immune system (cf. figure 8.10), MAdCAM-1 being largely absent from the genitourinary tract, lung, salivary and lacrimal gland, although it is present on vascular endothelium in the mammary gland.



immune system composed mostly of resident T-lymphocytes; similar set-ups appear to apply to the skin and to the liver. Nonclassical MHC antigens seem to play an important role in these specialized locales, with the MHC class I chain-related (MIC) family members MICA and MICB (cf. p. 78) implicated in the activation of human $\gamma\delta$ TCR IELs, and CD1 in antigen presentation to liver NK T-cells (cf. p. 102).

BONE MARROW CAN BE A MAJOR SITE OF ANTIBODY SYNTHESIS

A few days after a secondary response, activated memory B-cells migrate to the bone marrow where they mature into plasma cells (figure 8.12). The bone marrow is a major source of serum Ig, contributing up to 80% of the total Ig-secreting cells in the 100-week-

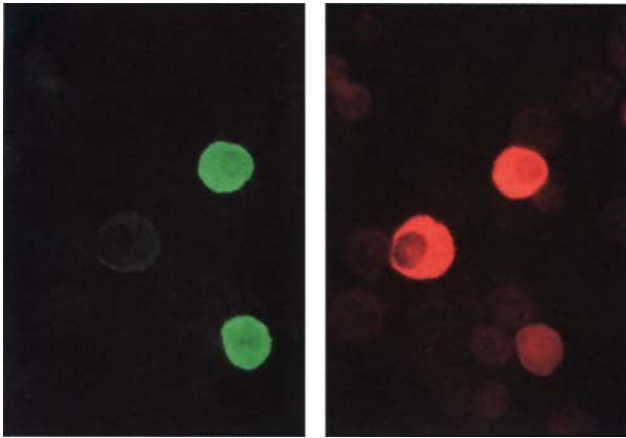


Figure 8.12. Plasma cells in human bone marrow. Cytospin preparation stained with rhodamine (orange) for IgA heavy chain and fluorescein (green) for lambda light chain. One cell is IgA.λ, another IgA.non-λ and the third is non-IgA.λ positive. (Photograph kindly supplied by Drs Benner, Hijmans and Haaijman.)

old mouse. The peripheral lymphoid tissue responds rapidly to antigen, but only for a relatively short time, whereas bone marrow starts slowly and gives a long-lasting massive production of antibody to antigens which repeatedly challenge the host.

THE ENJOYMENT OF PRIVILEGED SITES

Certain selected parts of the body, for example brain, anterior chamber of the eye and testis, have been designated **privileged immunological sites**, in the sense that antigens located within them do not provoke reactions against themselves. It has long been known, for example, that foreign corneal grafts can take up long-term residence, and a number of viruses have been expanded by repeated passage through animal brain.

Generally speaking, privileged sites are protected by rather strong blood–tissue barriers and low permeability to hydrophilic compounds and carrier-mediated transport systems. Functionally insignificant levels of complement reduce the threat of acute inflammatory reactions and unusually high concentrations of immunomodulators, such as IL-10 and transforming growth factor-β (TGFβ; cf. p. 179), endow macrophages with an immunosuppressive capacity. Immune privilege may also be maintained by Fas (CD95)-induced apoptosis of autoaggressive cells. Lesley Brent put it rather well: ‘It may be supposed that it is beneficial to the organism not to turn the anterior chamber or the cornea of the eye, or the brain, into an inflammatory battle-field, for the immunological response is sometimes more damaging than the antigen insult that provoked it.’

However, inflammatory reactions at the blood–tissue barrier can open the gates to invasion by immunological marauders—witness the inability of corneal grafts to take in the face of a local pre-existing inflammation.

THE HANDLING OF ANTIGEN

Where does antigen go when it enters the body? If it penetrates the tissues, it will tend to finish up in the draining lymph nodes. Antigens which are encountered in the upper respiratory tract or intestine are trapped by local MALT, whereas antigens in the blood provoke a reaction in the spleen. Macrophages in the liver will filter blood-borne antigens and degrade them without producing an immune response since they are not strategically placed with respect to lymphoid tissue.

Macrophages are general antigen-presenting cells

‘Classically’, it has always been recognized that antigens draining into lymphoid tissue are taken up by macrophages. The antigens are then partially, if not completely, broken down in the lysosomes; some may escape from the cell in a soluble form to be taken up by other antigen-presenting cells and a fraction may reappear at the surface, either as a large fragment or as a processed peptide associated with class II major histocompatibility molecules. Although resting, resident macrophages do not express MHC class II, antigens are usually encountered in the context of a microbial infectious agent which can induce the expression of class II by its adjuvant-like properties involving molecules such as bacterial lipopolysaccharide (LPS). The antigen-presenting cell must bear antigen on its surface for effective activation of lymphocytes and there is ample evidence that antigen-pulsed macrophages can stimulate specific T- and B-cells both *in vitro* and when injected back *in vivo*. Some antigens, such as polymeric carbohydrates like ficoll, cannot be degraded because the macrophages lack the enzymes required; in these instances, specialized macrophages in the marginal zone of the spleen (see figure 8.8d), or the lymph node subcapsular sinus, trap and present the antigen to B-cells directly, apparently without any processing or intervention from T-cells.

Interdigitating dendritic cells present antigen to T-lymphocytes

Notwithstanding this impressive account of the mighty macrophage in antigen presentation, there is

one function where it is seemingly deficient, namely the priming of naive lymphocytes. Animals which have been depleted of macrophages, by selective uptake of liposomes containing the drug dichloromethylene diphosphonate, are as good as their controls with intact macrophages in responding to T-dependent antigens. We must conclude that cells other than macrophages prime T-helper cells, and it is now generally accepted that these are the interdigitating dendritic cells (IDCs). These cells, which are of bone marrow origin, have the awesome capacity to process four times their own volume of extracellular fluid in 1 hour, thereby facilitating antigen capture and processing in their abundant intracellular MHC class II-rich compartments (MIIC; cf. p. 95).

The IDCs are the *crème de la crème* of the antigen-presenting cells and, if pulsed with antigen before injection into animals, usually produce stunning immune responses. In this connection, it is relevant to note that large numbers of these dendritic cells can be generated from peripheral blood by cultivation with granulocyte-macrophage colony-stimulating factor (GM-CSF) (cf. p. 179) to promote proliferation and IL-4 to suppress macrophage overgrowth; this means that it is perfectly feasible to contemplate their use for immunotherapy, e.g. pulsing autologous dendritic cells with the patient's tumor antigens and then reinjecting them to evoke an immune response.

Immature dendritic cells in the blood that are destined to become skin Langerhans' cells express cutaneous leukocyte antigen (CLA), directing their homing to skin via interaction with E-selectin on the relevant vascular endothelial cells just as occurs for cutaneous T-cells. The Langerhans' cells, and dendritic cells in other tissues, act as antigen sampling agents. They are only moderately phagocytic but display extremely active endocytosis. Receptors involved in antigen capture, including the mannose receptor and the immunoglobulin receptors Fc γ RII, Fc ϵ RI and Fc ϵ RII, are expressed at high levels. Unlike macrophages, however, they do not express appreciable amounts of the high affinity Fc γ RI. The expression of cell surface MHC class II, and of adhesion and costimulatory molecules, is low at this early stage of the dendritic cells' life. However, as they differentiate into fully fledged antigen-presenting cells, they decrease their phagocytic and endocytic activity, show reduced levels of molecules involved in antigen capture, but dramatically increase their MHC class II and CD1. Costimulatory molecules such as B7.1, B7.2 and CD40 are also upregulated at this stage, as are ICAM-1 and ICAM-2 which are thought to contribute to both the migratory and antigen-presenting properties of these

cells. Their expression of CD4 and the chemokine receptors CCR5 and CXCR4 (cf. table 10.3) means that they are attracted to and migrate into T-cell areas and incidentally become susceptible to infection by HIV (see p. 314).

There is evidence for different populations of IDCs, although this is still a somewhat shaky area. Two separate developmental pathways have been described, one which involves CD1a⁺ CD14⁻ cells with features of Langerhans' cells, and the other involving CD1a⁻ CD14⁺ cells that can differentiate into dermal dendritic cells that lack Langerhans' cell markers, such as Birbeck granules. Another subdivision that has been described involves the ability of murine CD8 α^- myeloid lineage dendritic cells to induce a Th2-type response, whilst CD8 α^+ lymphoid lineage dendritic cells activate Th1 differentiation (cf. p. 181). Recent data suggest a rather more complex situation, with dendritic cells arising from several different types of progenitor cells and the potential for functionally different types of dendritic cells to arise from the same progenitor, their function depending upon their maturation stage and the local cytokine environment.

What is clear is that, in addition to their antigen-presenting function, dendritic cells are important producers of chemokines. Lymph node IDCs that have matured from Langerhans' cells secrete T-cell-attracting chemokines, such as MDC (*macrophage-derived chemokine*) and TARC (*thymus- and activation-regulated chemokine*), with production localized to regions of the T-cell zone proximal to lymphoid follicles. MDC and perhaps TARC attract T-cells that are already to some extent activated. An apparently separate dendritic cell population constitutively expresses ELC (*EBI-1 ligand chemokine*) and SLC (*secondary lymphoid tissue chemokine*) in the T-cell zone and attracts naive T-cells. Thus a first phase of recruitment may occur involving naive T-cells, followed by a second wave of activated cells, thereby enhancing the possibility of encounter between antigen-bearing dendritic cells and their cognate antigen-specific T-cells. Once these cells get together, there are a number of receptor-ligand interactions involved in dendritic cell activation of T-cells aside from MHC-peptide recognition by the TCR. These include B7-CD28, CD40-CD40L, OX40L-OX40 (CD134) and TRANCE (TNF-related *activation-induced cytokine*)-TRANCE receptor recognition events. In addition to the T-cell zone, IDCs are also present in germinal centers, although not as prominently as the follicular dendritic cells to be discussed below. These germinal center IDCs may play a role in B-cell expansion and differentiation via their production of cytokines.

It is worth noting that, unlike macrophages, which in a sense are 'brutal microbe crunchers', the dendritic cells are not strongly phagocytic and they do need help from the macrophages in the preprocessing of particulate antigens, since these responses are completely abolished in macrophage-depleted animals.

To summarize, the scenario for T-cell priming appears to be as follows. Peripheral immature dendritic cells such as the Langerhans' cells (cf. figure 2.6f), which bind to skin keratinocytes through surface expression of E-cadherin, can pick up and process antigen. As maturation proceeds, they lose their E-cadherin and produce collagenase, presumably to facilitate their crossing of the basement membrane. They then travel as 'veiled' cells in the lymph (figure 8.13b) before settling down as IDCs in the paracortical T-cell zone of the draining lymph node (figure 8.13a). There, maturation is completed (figure 8.14), the IDC delivers the antigen with costimulatory signals for potent stimulation of naive and subsequently of activated, specific T-cells, which take advantage of the large surface area to bind to the MHC-peptide complex on the IDC membrane.

We will meet IDCs again in Chapter 12 when we discuss their central role within the thymus where they present self-peptides to developing autoreactive T-cells and trigger their apoptotic execution (known more gently as 'clonal deletion'; cf. p. 231).

Follicular dendritic cells stimulate B-cells in germinal centers

Another type of cell with dendritic morphology, but this time of mesenchymal origin, exists within lymphoid tissues and is referred to as the **follicular dendritic cell (FDC)**. These are nonphagocytic and lack lysosomes but have very elongated processes which can make contact with numerous lymphocytes present within the germinal centers of secondary follicles. Their possession of FcγRII, FcεRII and complement receptors enables them to trap complexed antigen very efficiently and hold it in its native form on their surface for extended periods, in keeping with the memory function of secondary follicles. This would explain the fact that secondary antibody responses can be boosted by quite small amounts of immunogen which become bound by circulating antibody and then fix C3. Evidence for this notion may be derived from animals effectively depleted of complement by injection of cobra venom factor which contains the reptilian equivalent of C3b. This fires the alternative complement pathway by forming a complex with factor B but, because of its insensitivity to the mammalian C3 inactivator, it persists long enough to discharge the feedback loop to exhaustion and deplete C3 completely. Such mice can neither localize antigen-antibody complexes on their

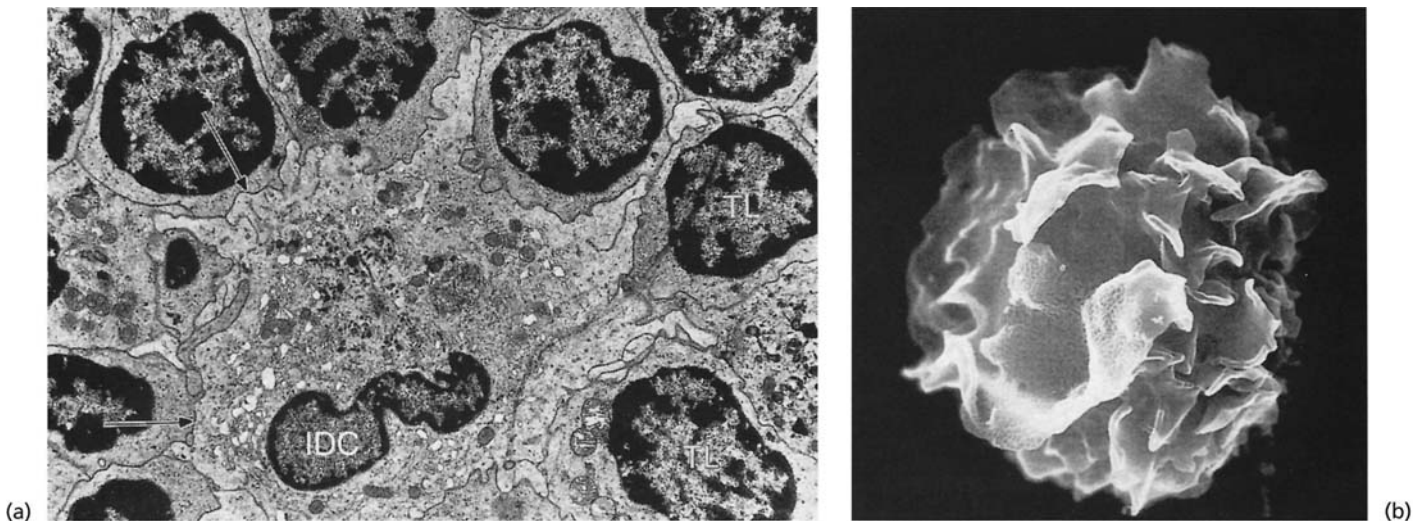


Figure 8.13. Dendritic antigen-presenting cell. (a) Interdigitating dendritic cell (IDC) in the thymus-dependent area of the rat lymph node. This is thought to be an antigen-presenting cell derived from the Langerhans' cell in the skin and dendritic cells in other tissues, which travels to the node in the afferent lymph as a 'veiled' cell bearing antigen on its profuse surface processes. Intimate contacts are made with the surface membranes (arrows) of the surrounding T-lymphocytes (TL). The cytoplasm of the IDC contains relatively few organelles and does not show Birbeck granules (racket-shaped cytoplasmic organelles, characteristic of the Langerhans' cell), although some IDCs in lymph nodes do possess these granules, perhaps after antigenic stimulation ($\times 2000$). (b) Scanning electron micrograph of a veiled cell. In contrast with these dendritic cells which present antigen to T-cells, the follicular dendritic cells in germinal centers stimulate B-cells. ((a) Reproduced with permission of the authors and publishers from Kamperdijk E.W.A., Hoefsmit E.Ch.H., Drexhage H.A. & Balfour B.H. (1980) In Van Furth R. (ed.) *Mononuclear Phagocytes*, 3rd edn. Rijhoff Publishers, The Hague. (b) Courtesy of Dr G.G. MacPherson.)

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Figure 8.14. Migration and maturation of interdigitating dendritic cells. The precursors of the IDCs are derived from bone marrow stem cells. They travel via the blood to nonlymphoid tissues. These immature IDCs, e.g. Langerhans' cells in skin, are specialized for antigen uptake. Subsequently they travel via the afferent lymphatics as veiled cells (cf. figure 8.13b) to take up residence within secondary lymphoid tissues (cf. figure 8.13a) where they express high levels of MHC class II and costimulatory molecules such as B7. These cells are highly specialized for the activation of naive T-cells.

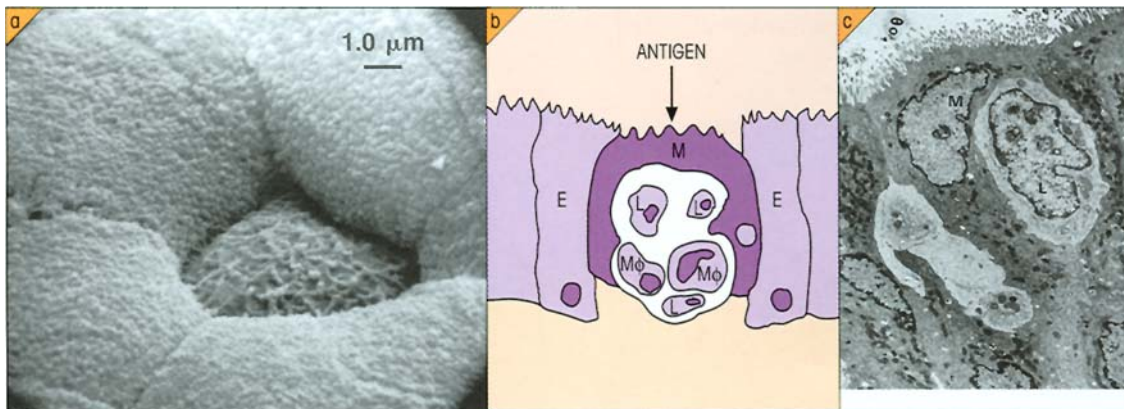
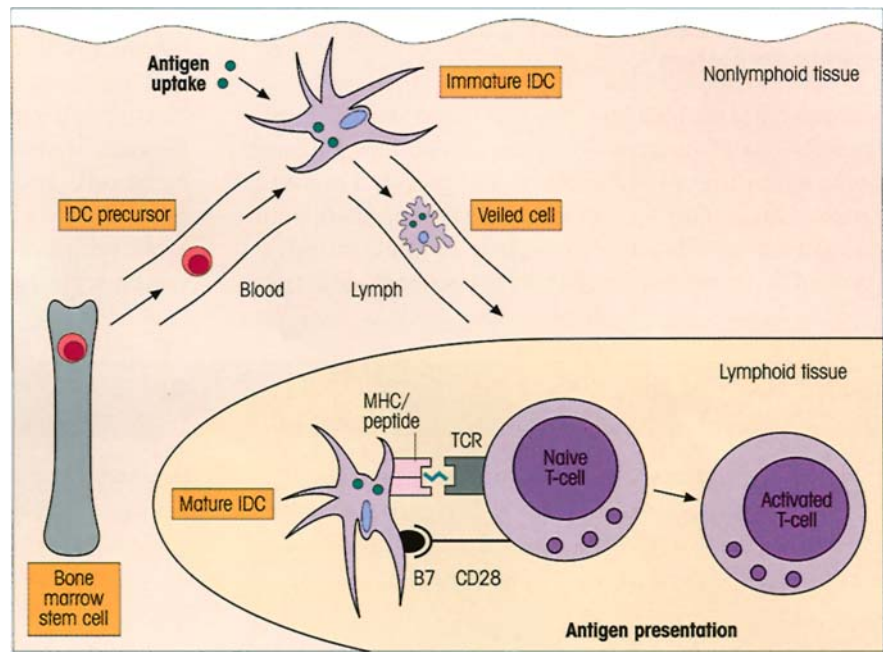


Figure 8.15. M-cell within Peyer's patch epithelium. (a) Scanning electron micrograph of the surface of the Peyer's patch epithelium. The antigen-sampling M-cell in the center is surrounded by absorptive enterocytes covered by closely packed, regular microvilli. Note the irregular and short microfolds of the M-cell. (Reproduced with permission of the authors and publishers from Kato T. & Owen R.L. (1999) In Ogra R. *et al.* (eds) *Mucosal Immunology*, 2nd edn. Academic Press, San Diego.) (b) After uptake and transcellular transport by the M-cell (M), antigen is processed by macrophages and thence by IDCs which present antigen to T-cells in Peyer's patches and mesenteric

lymph nodes. E, enterocyte; L, lymphocyte; Mφ, macrophage. (c) Electron photomicrograph of an M-cell (M in nucleus) with adjacent lymphocyte (L in nucleus). Note the flanking epithelial cells are both absorptive enterocytes with a typical brush border. In some cases, proteases on the surface of the M-cells modify the pathogen so that it can adhere and be taken up. Pathogenic *Salmonella* can invade and destroy M-cells, making a hole through which other bacteria can invade the underlying tissue. (Lead citrate and uranyl acetate, ×1600.) (b) Based on Sminia T. & Kraal G. (1998) In Delves P.J. & Roitt I.M. (eds) *Encyclopedia of Immunology*, 2nd edn, p. 188. Academic Press, London.)

follicular dendritic cells, nor generate memory B-cells in response to T-dependent antigens (see p. 171).

Classically, a secondary response would be initiated at the T-helper level by antigen, alone or as a complex, being taken up by IDCs and macrophages. However, the capture of immune complexes on the surface of FDCs opens up an alternative pathway. One to three days after secondary challenge, the filamentous den-

drites on the follicular cells, to which the immune complexes are bound, form into beads which break off as structures called 'icosomes' (immune complex-coated bodies). These bind to germinal center B-cells which then endocytose and process the antigen for presentation by the B-cell MHC class II, and subsequent stimulation of T-helper cells to kick off the secondary response.

M-cells provide the gateway to the mucosal lymphoid system

The mucosal surface is in the front line facing an unfriendly sea of microbes and, generally, antigens are excluded by the epithelium with its tight junctions and mucous layer. Gut lymphoid tissue is separated from the lumen by a layer of columnar epithelium interspersed with microfold (M)-cells (figure 8.15a); spe-

cialized antigen-transporting cells with short, irregular microvillae, strong nonspecific esterase activity and no MHC class II. They overlay intraepithelial lymphocytes and macrophages (figure 8.15b and c). Foreign material, including bacteria, is taken up by M-cells and passed on to the underlying antigen-presenting cells which, in turn, migrate to the local lymphoid tissue to activate the appropriate lymphocytes.

SUMMARY

The surface markers of cells in the immune system

- Individual surface molecules are assigned a cluster of differentiation (CD) number defined by a cluster of monoclonal antibodies reacting with that molecule.

Organized lymphoid tissue

- The complexity of immune responses is catered for by a sophisticated structure.
- Lymph nodes filter and screen lymph flowing from the body tissues while spleen filters the blood.
- B- and T-cell areas are separated.
- B-cell structures appear in the lymph node cortex as primary follicles which become secondary follicles with germinal centers after antigen stimulation.
- Germinal centers with their meshwork of follicular dendritic cells expand B-cell blasts produced by secondary antigen challenge and direct their differentiation into memory cells and antibody-forming plasma cells.

Mucosal-associated lymphoid tissue

- Lymphoid tissue guarding the gastrointestinal tract is unencapsulated and somewhat structured (tonsils, Peyer's patches, appendix), or present as diffuse cellular collections in the lamina propria. Intraepithelial lymphocytes are mostly T-cells and include some novel subsets, e.g. CD8 $\alpha\alpha$ -bearing cells which use nonclassical MHC molecules as restriction elements for antigen presentation.
- Together with the subepithelial accumulations of cells lining the mucosal surfaces of the respiratory and genitourinary tracts, this lymphoid tissue forms the 'secretory immune system' which bathes the surface with protective IgA antibodies.

Other sites

- Bone marrow is a major site of antibody production.
- The brain, anterior chamber of the eye and testis are privileged sites in which antigens can be safely sequestered.

Lymphocyte traffic

- Lymphocyte recirculation between the blood and lymphoid tissues is guided by specialized homing receptors on the surface of the high-walled endothelium of the postcapillary venules.
- Lymphocytes are tethered and then roll along the surface of the selected endothelial cells through interactions between selectins and integrins and their respective ligands. Flattening of the lymphocyte and transmigration across the endothelial cell follow LFA-1 activation.
- Entry of memory T-cells into sites of inflammation is facilitated by upregulation of integrin molecules (VLA-4 and LFA-1) on the lymphocyte and corresponding binding ligands on the vascular endothelium (VCAM-1 and ICAM-1/2 respectively).

The handling of antigen

- Macrophages are general antigen-presenting cells for primed lymphocytes but cannot stimulate naive T-cells.
- This is effected by dendritic cells of hematopoietic origin which process antigen, migrate to the draining lymph node and settle down as interdigitating dendritic cells. They can present antigen-derived peptides to naive T-cells, thereby powerfully initiating primary T-cell responses.
- Follicular dendritic cells in germinal centers bind immune complexes to their surface through Ig and C3b receptors. The complexes are long-lived and provide a sustained source of antigenic stimulation for B-cells.
- Specialized antigen-transporting M-cells provide the gateway for antigens to the mucosal lymphoid tissue.

See the accompanying website (www.roitt.com) for multiple choice questions.